

Introduction to Human Biology Laboratory Workbook

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Preface



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This manual includes original content by George Mateja, Professor of Biology at Community College of Baltimore County. Its creation also contains a variety of remixed and revised content from other resources. These sources are noted in the attributions section for each lab.

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	Topics related to writing in lab
	Front Matter 01: Scientific Writing, Citation, and Using Credible Sources
	Front Matter 02: Prelab Preparation and Responsible Use of Ai
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1	Lab 01: Laboratory Safety and Measurement
2	Lab 02: Use of the Microscope
3	Lab 03: Molecules, Water, and pH; Bufferin Variation
4	Lab 04: Chemical Composition of Cells - VM (take home handout-coupled with lab 04) Molecular Structure of Biological Molecules
5	Lab 05: Diffusion and Osmosis
6	Lab 06: Enzyme activity (Turnip Peroxidase)
7	Lab 07: Practical 1
8	Lab 08: Respiration and Yeast Fermentation
9	Lab 09: Photosynthesis & Respiration
10	Lab 10: Mitosis & Meiosis (VM - not sent)
11	Lab 11: Mendelian Genetics of Corn with Inherited Human Traits
12	Lab 12: Mendelian Genetics Problems with Genetics Handout.
13	Lab 13: (Break; floats depending on semester)
14	Lab 14: Poster presentation & Practical Review
15	Lab 15: Practical 2
16	Finals week – No Labs

Scientific Writing, Citations, and Using Credible Sources

Student Pre-Lab Questions

1. What makes a question scientific?
2. Why must credible sources be used in biology?
3. What is a primary scientific source?
4. What is plagiarism?
5. When must a source be cited?
6. What is included in an in-text citation?
7. Which section explains why a study was conducted?

Overview

Biology is a scientific discipline grounded in observation, experimentation, and evidence-based reasoning. Scientific knowledge does not exist in isolation; it builds upon the work of other researchers. As a result, scientists must evaluate credible sources, cite prior work properly, and communicate findings clearly and accurately.

In this course, you will learn how to:

- Ask scientific, testable questions
- Identify and evaluate credible biological sources
- Properly cite scientific literature
- Communicate findings using accepted formats of scientific writing

Together, these skills form the foundation of scientific literacy and integrity.

Learning Objectives

By the end of this chapter, students will be able to:

1. Distinguish between scientific and non-scientific questions
2. Identify credible sources appropriate for biological research
3. Differentiate between primary, secondary, popular, and scholarly sources
4. Evaluate source credibility using objective criteria (CRAAP test)
5. Explain the importance of citation in scientific writing
6. Define plagiarism and identify examples of academic dishonesty
7. Apply proper in-text citation formats (APA or CSE style)
8. Describe the structure and purpose of sections in a scientific paper
9. Distinguish between Results and Discussion sections
10. Use writing practices that improve clarity, accuracy, and credibility

Scientific Questions and the Role of Evidence

Science focuses on questions that can be tested through observation, measurement, and experimentation.

Scientific Questions

- Can be investigated empirically
- Rely on measurable data
- Are testable and repeatable

Examples:

- What temperature maximizes bacterial growth?
- Does a specific drug reduce disease symptoms?

Non-Scientific Questions

- Cannot be tested using scientific methods
- Often involve opinions, beliefs, or ethics

Examples:

- What is the best genre of music?
- Should cloning be allowed?

Only scientific questions can be addressed in laboratory investigations.

Using Credible Sources in Biology

Before conducting research, scientists review existing knowledge. Using credible sources ensures accuracy and reliability.

Types of Scientific Sources

Primary Sources

- Original research articles
- Present new data and experimental findings
- Peer-reviewed and written by experts

Secondary Sources

- Summarize or interpret primary research
- Examples: textbooks, review articles

Popular vs. Scholarly Sources

The following table compares popular and scholarly sources, highlighting key differences in audience, review process, and reliability in scientific research.

Table 1. Comparison of Popular and Scholarly Sources in Scientific Research

This table compares popular and scholarly sources across several characteristics. Popular sources are written for a general audience, are not peer-reviewed, contain few or no citations, use simplified language, and include examples such as blogs and magazines. Scholarly sources are written for an expert audience, are peer-reviewed, include extensive references, use technical language, and include examples such as scientific journals like *Science* and *Nature*.

Popular Sources	Scholarly Sources
General audience	Expert audience
Not peer-reviewed	Peer-reviewed
Few or no citations	Extensive references
Simplified language	Technical language
Examples: blogs, magazines	Examples: <i>Science</i> , <i>Nature</i>

Scientific writing relies primarily on **scholarly sources**.

Evaluating Credibility: The CRAAP Test

To assess sources, use the CRAAP criteria:

- **Currency** – Is the information recent?
- **Reliability** – Is it from a trusted organization or journal?
- **Authority** – Is the author qualified?
- **Accuracy** – Is it supported by evidence?
- **Purpose** – Is the content objective or biased?

Low-quality sources should not be used in scientific work.

Citing Sources and Avoiding Plagiarism

Why Citation Matters

Citations:

- Distinguish your ideas from others
- Credit original authors
- Allow verification of information
- Maintain academic integrity

What Is Plagiarism?

Plagiarism is presenting another person's work as your own.

Examples:

- Copying text without citation
- Paraphrasing without credit
- Submitting someone else's work

Avoiding Plagiarism

Plagiarism is the act of presenting another person's work, ideas, words, or creative output as your own without giving proper credit to the original source. This can include copying text directly, paraphrasing someone else's ideas without acknowledgment, or submitting work that was created by someone else, such as a classmate or an online source. Plagiarism is considered a serious academic and ethical violation because it misrepresents authorship, undermines intellectual honesty, and deprives the original creator of recognition for their work.

- Use your own words
- Cite all non-original ideas
- Keep accurate research notes
- Use quotes sparingly in science

Citation Styles in Science

Scientific disciplines use standardized formats such as:

- **APA (Author–Year)**
- **CSE (Name–Year commonly used in biology)**

In-Text Citation Example

- (Smith, 2020)

Reference List

- Includes only sources cited in the paper
- Organized alphabetically

Consistency in formatting is essential.

Writing for Science

Scientific writing emphasizes **clarity**, **precision**, and **conciseness** to ensure that ideas and findings are communicated effectively. Clarity means presenting information in a straightforward and understandable manner, avoiding ambiguity or unnecessary complexity. Precision involves using specific, accurate language and terminology so that the meaning is exact and scientifically correct. Conciseness requires eliminating unnecessary words or repetition, allowing the reader to quickly grasp the essential information without distraction. Together, these principles help make scientific communication efficient, credible, and accessible to other researchers.

Structure of a Scientific Paper

The structure of a scientific paper provides a standardized format that allows researchers to communicate findings clearly and logically. Each section has a specific purpose: the Introduction presents background and the research question, the Methods describe how the study was conducted, the Results report the data, and the Discussion interprets the findings. This organized approach helps readers quickly understand the study and evaluate its validity.

Title

- Concise and descriptive

Abstract

- Summary of purpose, methods, results, and conclusions
- Written last

Introduction

- Provides background and context
- States research question or hypothesis
- Explains importance of the study

Methods

- Describes how the study was conducted
- Allows replication
- Includes materials and procedures

Results

- Presents data and findings
- Includes figures and tables
- No interpretation

Discussion

- Interprets results
- Connects findings to existing research
- Explains significance and limitations

Literature Cited

- Lists all referenced sources

General Writing Practices

- Be concise and precise
- Use appropriate scientific terminology
- Prefer active voice when appropriate
- Use past tense for results, present tense for general knowledge
- Avoid ambiguity and unnecessary detail
- Treat “data” as plural
- Italicize scientific names (*Homo sapiens*)
- Revise and proofread carefully

Summary and Application to Laboratory Work

This chapter introduces the foundational skills required for success in biology, emphasizing the importance of scientific thinking, credible research, proper citation, and effective communication. Biology relies on evidence-based investigation, meaning that students must be able to ask testable scientific questions, evaluate reliable sources, and interpret data accurately. Understanding how to distinguish between scientific and non-scientific questions ensures that laboratory work remains focused on measurable and reproducible outcomes.

A key component of scientific work is the use of **credible sources**, particularly scholarly and peer-reviewed materials. Students learn to differentiate between primary and secondary sources and to evaluate credibility using criteria such as the CRAAP test. This ensures that conclusions are based on accurate, unbiased, and scientifically valid information.

Equally important is the ethical use of information. Proper **citation practices** allow students to give credit to original researchers, avoid plagiarism, and strengthen the credibility of their work. Using consistent citation formats such as APA or CSE helps maintain professional standards in scientific communication.

The chapter also introduces **scientific writing**, which prioritizes clarity, precision, and conciseness. Students learn the standard structure of a scientific paper—including Introduction, Methods, Results, and Discussion—and the purpose of each section. This structure ensures that experiments can be clearly understood, evaluated, and replicated.

Application to a Pre-Lab or Laboratory Report

In a **pre-lab**, students apply these skills by:

- Formulating testable scientific questions or hypotheses
- Researching background information using credible sources
- Properly citing references used to justify the experiment

In a **laboratory report**, students:

- Use the **Introduction** to explain the scientific question and support it with cited research
- Describe procedures clearly in the **Methods** to allow replication
- Present data objectively in the **Results** section
- Interpret findings in the **Discussion**, connecting results to scientific literature
- Include a properly formatted **Literature Cited** section

Overall, these skills ensure that laboratory work is not only accurate and ethical but also aligned with the standards of the scientific community.

Student Post-Lab Questions

1. How do citations improve scientific credibility?
2. What distinguishes scholarly from popular sources?
3. Give one example of plagiarism and explain why it is incorrect.
4. Why must Methods be replicable?
5. Why is interpretation excluded from Results?

6. How does Discussion differ from Results?
7. How do credible sources strengthen scientific conclusions?

Appendix: Scientific Writing, Citations, and Credible Sources Terms

Term	Definition
Abstract	A brief summary of a scientific paper including purpose, methods, results, and conclusions.
Accuracy	The correctness of information, supported by evidence and reliable sources.
APA Style	A citation format that uses author and date for referencing sources.
Authority	The credibility and qualifications of an author or source.
Bias	A tendency to present information in a non-objective or unfair way.
Citation	A reference to a source used to give credit and allow verification.
Clarity	The quality of being easy to understand in writing.
Conciseness	Expressing ideas clearly using as few words as necessary.
CRAAP Test	A tool used to evaluate sources based on Currency, Reliability, Authority, Accuracy, and Purpose.
Credible Source	A reliable and trustworthy source of information supported by evidence and expertise.
CSE Style	A citation format commonly used in biology that emphasizes author and year.
Discussion	A section of a scientific paper that interprets results and explains their significance.
Empirical	Based on observation or experiment rather than theory alone.
Introduction	The section of a paper that provides background and states the research question or hypothesis.
Literature Cited	A list of all sources referenced in a scientific paper.

Term	Definition
Methods	The section describing how an experiment was conducted.
Peer Review	The evaluation of scientific work by experts before publication.
Plagiarism	Using another person's work or ideas without giving proper credit.
Precision	Using exact and accurate language in scientific writing.
Primary Source	An original research article presenting new data.
Purpose	The intended goal or objective of a source or study.
Reliability	The consistency and dependability of a source.
Results	The section that presents data without interpretation.
Scholarly Source	A peer-reviewed source written for a scientific audience.
Scientific Question	A question that can be tested through observation and experimentation.
Secondary Source	A source that summarizes or interprets primary research.
Significance	The importance or meaning of findings in a scientific context.
Source Evaluation	The process of determining the credibility of a source.
Testable	Able to be examined through scientific experiment or observation.

Attribution Note

This chapter is adapted and synthesized from openly licensed instructional materials on scientific writing, citation practices, and evaluating sources, including OpenStax Biology and university library resources. Adapted for introductory biology laboratory instruction.

Pre-Lab Preparation and Responsible Use of AI

Student Information (Required for Submission)

- Name: _____
- Date: _____
- Lab Section: _____
- Lab Title: _____

Overview of Pre-Lab Preparation

Preparing for your laboratory session is an important part of your success in this course. Before attending lab, you are expected to carefully review all assigned materials, including the lab manual, background readings, and any additional instructions provided by your instructor.

Effective reading requires **active engagement**. This means you should take your time, avoid skimming, and review the material more than once if necessary to fully understand the concepts and procedures. Thoughtful preparation will help you:

- Work more efficiently during the lab
- Make accurate observations
- Better understand the purpose of the experiment

Important Note:

Not all instructors require a pre-lab assignment. Requirements may vary by faculty member, so it is your responsibility to check your course instructions or confirm expectations with your instructor before completing this work.

Responsible Use of Artificial Intelligence (AI)

This course recognizes that students may use AI tools (such as ChatGPT, Claude, or Copilot) to support learning. When used appropriately, AI can be helpful for:

- Generating ideas or study questions
- Clarifying difficult concepts
- Improving grammar and writing clarity

However, AI is **not a replacement for your own thinking or scientific understanding**.

Key Expectations

- AI is **not a reliable scientific source**
- AI responses may be **incorrect, incomplete, or misleading**
- You are responsible for **verifying all information** and completing your own work

AI Guidelines for This Lab

You are expected to:

- Use AI only as a **support tool**, not as a source of answers
- Verify all information using **textbooks or approved scientific sources**
- Write all responses in **your own words**
- Take full responsibility for the **accuracy of your work**

You are not allowed to:

- Copy or paste AI-generated content into your assignment
- Submit AI-generated answers as your own work
- Use AI instead of reading assigned lab materials

If you use AI to help organize ideas or revise writing, you may acknowledge its use according to your instructor's guidelines. All scientific content must come from credible sources.

Grading

Each section of the pre-lab assignment is worth **5 points**.

Pre-Lab Assignment Sections

Section 1: Essential Ideas of the Lab

Identify the main scientific concept or procedure you will study.

Your response should address:

(a) What is already known about this topic? Include:

- Key concepts and definitions
- Important discoveries or developments
- Any gaps or unanswered questions

(b) Why was this study performed?

- Explain the purpose of the lab using the title and introduction

Requirements:

- Write in your own words
- Use credible sources (e.g., textbooks, lab manual)
- Cite sources in APA format

Section 2: Lab Objectives

Describe what you will *do* during the lab.

Examples of objective actions:

- Observe
- Measure
- Analyze
- Determine

Requirements:

- Use action-based statements (e.g., “To observe...”)
- Focus on the procedures you will perform

Section 3: Purpose of the Lab

Explain how lab activities support your understanding of the scientific concept.

This section connects:

- What you are doing
- What you are learning

Requirement:

- Clearly explain this relationship in your own words

Section 4: Hypothesis

(Required for experimental labs only: Labs 2, 4, 6, 7, 8, and 9)

Develop a hypothesis (an educated prediction).

Steps:

- Identify the independent variable (what you change)
- Identify the dependent variable (what you measure)
- Predict the relationship between them

Format:

If [independent variable changes], then [dependent variable will respond], because of [scientific reasoning].

Requirements:

- Base your hypothesis on scientific knowledge
- Clearly explain your reasoning
- Cite any supporting information

Section 5: Questions

Write at least **five thoughtful questions** about the lab.

Your questions may focus on:

- Concepts you find confusing
- Steps in the procedure
- Expected results

Section 6: References Cited

List all sources in **APA (7th edition) format**.

CCBC APA Guide:

<https://libraryguides.ccbcmd.edu/Citations/apa7>

Requirements:

- Include all sources used
- Use proper in-text citations
- Provide a complete reference list

Pre-Lab Checklist (Before Submission)

I carefully read all assigned materials

I confirmed whether this pre-lab is required for my section

I used AI appropriately (if applicable)

I verified all information using reliable sources

I wrote all responses in my own words

I explained how the lab connects to the concept

I included at least five questions

I formatted all references in APA style

Final Note to Students

This pre-lab assignment is designed to help you prepare for the laboratory experience and improve your understanding of scientific concepts. Students who complete the pre-lab thoughtfully are typically more confident, efficient, and successful in the lab.

Remember: success in this course depends on **preparation, critical thinking, and academic integrity**. Responsible use of AI and independent learning are essential skills for both this class and future scientific work.

Appendix: Pre-Lab Preparation and Responsible Use of AI Terms

Term	Definition
Academic Integrity	Honest and ethical behavior in academic work.
Action Verb	A word that describes an action, used in lab objectives (e.g., observe, measure).
APA Format	A standardized citation style used to reference sources in scientific writing.
Artificial Intelligence (AI)	Computer systems that can generate text, analyze information, or assist with learning tasks.
Checklist	A list used to ensure all required steps or components are completed.
Citation	Giving credit to a source of information.
Concept	A main idea or principle studied in a lab or lesson.
Credible Source	A reliable and trustworthy source of scientific information.
Dependent Variable	The variable that is measured or observed in an experiment.
Experimental Design	The plan for conducting an experiment and testing a hypothesis.
Hypothesis	An educated prediction that can be tested through experimentation.
Independent Variable	The variable that is changed or manipulated in an experiment.
Lab Objective	A statement describing what students will do during a lab.

Term	Definition
Paraphrasing	Rewriting information in your own words while retaining meaning.
Plagiarism	Using another person's words or ideas without proper credit.
Pre-Lab Preparation	The process of reviewing material before conducting an experiment.
Procedure	Step-by-step instructions for completing a lab experiment.
Purpose of Lab	An explanation of how the lab connects to scientific concepts.
Reference List	A complete list of sources used in an assignment.
Responsible AI Use	Using AI tools ethically to support learning without replacing original work.
Scientific Reasoning	Using logic and evidence to explain scientific ideas.
Source Evaluation	The process of determining whether a source is reliable and accurate.
Study Question	A question used to guide understanding before or during a lab.
Support Tool	A resource used to assist learning without replacing student work.
Verification	The process of checking information for accuracy.
Writing Clarity	The ability to express ideas clearly and effectively.

1 Laboratory Safety and Measurement

1.1 Introduction: Laboratory Safety and Measurement

Welcome to your first biology lab! Before you begin conducting experiments, it is essential to learn how to work safely and accurately in a laboratory setting. This lab will introduce you to the basic skills and habits that scientists use every day.

In the first part of this lab, you will learn about **laboratory safety**. Even in an introductory biology lab, there are potential hazards such as chemicals, glassware, biological materials, and heat sources. Understanding how to recognize these risks—and how to respond to them—helps protecting you, your classmates, and your experiments. Following safety rules also ensure that your results are reliable and free from contamination. These safety practices are not only important in class but are also essential in real-world careers such as healthcare, research, and biotechnology.

In the second part of this lab, you will focus on **scientific measurement**, which is a key skill in all areas of science. You will learn how to use the **metric (SI) system**, the standard system used by scientists worldwide. You will practice measuring length, mass, and volume using common laboratory tools such as rulers, balances, graduated cylinders, and pipettes. You will also learn how to convert between units, record data correctly, and organize your results.

Finally, you will explore how scientists evaluate data by learning the difference between **accuracy** (how close a measurement is to the true value) and **precision** (how consistent repeated measurements are). You will also practice calculating percent error and displaying your data using graphs.

Together, these skills—working safely, measuring carefully, and recording data accurately—form the foundation for all future lab work. Mastering them now will help you succeed in this course and prepare you for more advanced scientific investigations.

Laboratory Safety and Measurement Objectives

By the end of this laboratory unit, students will be able to:

A. Laboratory Safety Knowledge

1. Identify common laboratory hazards like chemical, biological, physical, and equipment.

2. Explain the importance of laboratory safety in preventing injuries, ensuring valid results, and preparing for workplace environments.
3. Describe proper laboratory attire and personal protective equipment (PPE) and explain how each protects against specific hazards.

B. Laboratory Safety Skills

1. Demonstrate safe laboratory behaviors, including proper handling of materials, maintaining clean workspaces, and following instructions.
2. Apply correct emergency procedures, including responding to chemical spills and eye or skin exposure.
3. Properly dispose of laboratory waste, including glass, biological materials, and chemical substances.

C. Scientific Measurement Skills

1. Identify and use standard SI (metric) units for length, mass, volume, and temperature.
2. Perform accurate metric unit conversions using appropriate prefixes (kilo-, centi-, milli-, micro-).
3. Use laboratory instruments correctly, including rulers, balances, graduated cylinders, and pipettes.

D. Data and Scientific Reasoning

1. Record and report data accurately, including correct units, labels, and appropriate precision.
2. Differentiate between accuracy and precision and evaluate measurements based on these concepts.
3. Analyze measurement error and identify potential sources of error in laboratory procedures.

E. Professional and Scientific Practice

1. Demonstrate responsibility and accountability in a laboratory setting by following safety guidelines and maintaining appropriate conduct.
2. Explain how laboratory safety practices translate to real-world careers in healthcare, research, and biotechnology.

1.2 Introduction to Laboratory Safety

Purpose

Laboratory safety is essential for protecting yourself, your classmates, and your experiments. Even in introductory biology labs, hazards such as chemicals, glassware, biological materials, and heat sources are present.

Why This Matters (Lab and Workplace)

- Prevents injuries such as burns, cuts, and exposure to harmful materials
- Ensures accurate experimental results (no contamination or errors)
- Builds professional habits required in healthcare, research, and laboratory jobs

Example

If you contaminate a microbial sample by touching it without gloves, results become invalid and unusable. In a workplace lab, this could lead to incorrect diagnoses or product failure.

1.3 Pre-Lab Activity 1.1: Questions

Multiple Choice

1. Why is lab safety important?
 - A. Saves time
 - B. Prevents injury and ensures accurate results
 - C. Reduces workload
 - D. Makes lab easier
2. Which footwear is acceptable?
 - A. Sandals
 - B. Closed-toe shoes
 - C. Flip-flops
 - D. Open-toe shoes
3. What is a common lab hazard?
 - A. Food
 - B. Chemicals
 - C. Chairs
 - D. Paper

Short Answer

4. Why must hair be tied back in the lab?

5. What could happen if you do not follow safety rules?

General Safety Review

1.4 Section 1.1: Common Laboratory Hazards

Introduction

Even basic biology labs contain hazards that can cause injury if proper precautions are not followed.

Types of Hazards

- Chemicals (toxins, irritants)
- Sharp objects (glassware, pipettes)
- Biological materials (specimens, fluids)
- Heat and electrical equipment

Examples

- Breaking a glass slide can cut your hand
- Hot plates can cause burns if touched incorrectly
- Preserved specimens may contain irritating chemicals

1.5 Section 1.2: Proper Clothing and Personal Preparation

Introduction

Clothing and preparation protect you before accidents occur.

Required Clothing

- Closed-toe, closed-heel shoes
- Long pants covering legs
- Shirt covering torso and shoulders
- Hair tied back

Not Allowed

- Sandals, flip-flops, open shoes
- Shorts or exposed legs
- Loose sleeves or dangling items

Examples

- A chemical spill on exposed skin can cause irritation or burns
- Loose hair may catch fire or fall into chemicals

1.6 Section 1.3

Introduction

PPE is your direct barrier against hazards.

Required PPE

- Safety goggles (ANSI-approved)
- Gloves (when handling biological or chemical materials)
- Lab coat (recommended)

Examples

- Goggles prevent chemicals from entering eyes
- Gloves prevent exposure to biological materials

1.7 Section 1.4: Safe Lab Behavior

Introduction

Safe behavior reduces accidents for everyone in the lab.

Rules

- No eating or drinking
- No running or horseplay
- Keep hands away from face
- Treat all materials as potentially hazardous

Examples

- Eating in the lab could expose you to chemicals
- Running could cause spills or broken glass

1.8 Section 1.5: Emergency Procedures

Introduction

Knowing what to do immediately can prevent severe injury.

Chemical Splash (Face with Goggles)

- Do NOT remove goggles
- Go to eyewash station immediately
- Flush for at least 15 minutes
- Notify instructor

Chemical Splash (Eyes)

- Use eyewash immediately
- Hold eyes open
- Flush for at least 15 minutes

Examples

- Rapid flushing can prevent permanent eye damage
- Delaying response increases injury severity

1.9 Section 1.6: Waste Disposal and Cleanup

Introduction

Proper disposal protects people and the environment.

Guidelines

- Dispose of biohazard materials properly
- Use broken glass containers

- Never leave spills unattended
- Clean lab benches before leaving

Example

Improperly discarded glass can injure custodial staff.

1.10 Section 1. 7: Hand Washing and Hygiene

Introduction

Hygiene prevents contamination and illness.

Key Rule

- Wash hands for at least 30 seconds before leaving lab

Example

Failure to wash hands may spread biological contaminants outside the lab.

Measurement and the Metric System

Learning Objectives

By the end of this lab, students will be able to:

1. Identify the common International System of Units (SI) used in biology.
2. Explain why the SI system is preferred over the imperial system in science.
3. Perform conversions between metric units using the following prefixes: k-, d-, c-, m- and μ -.
4. Perform measurements of length, mass, and volume using proper scientific equipment.
5. Record and report experimental data using correct metric units.
6. Distinguish between accuracy and precision in laboratory measurements.
7. Record data on charts to include proper units and labels.

Prelab Activity 1.1:

Students should complete these questions before coming to the lab.

1. What is the SI base unit for length, mass, and volume commonly used in biology?
2. How many milliliters are in one liter?

3. What metric prefix stands for one thousandth of a base unit?
4. Why is the metric system easier to use than the imperial system in science?
5. Which unit would be most appropriate for measuring the size of a cell, and why?
6. What measuring device would you use to measure mass in a laboratory?
7. Explain the difference between accuracy and precision.

Background:

Scientists use scientific methods to understand the natural world. An important aspect of the scientific process is collecting and analyzing data from observations and experiments. Two methods of measuring and recording data have been commonly used - the Imperial and the Metric System. The Imperial System uses units such as inches and pounds while the Metric System, a decimal-based system, uses units such as meters, liters, and grams. Because of its ease of use, the metric system is more widely used within the modern scientific community. The universal “language” of modern science is based on the metric system and learning to use it is the main purpose of this lab.

Part 1: Introduction to Metric System

Work conducted in science laboratories requires a functional understanding of the metric system, which serves as the standard system of measurement in scientific research. The metric system employs standardized, decimal-based units to measure length, mass, and volume, ensuring that measurements are precise, reproducible, and universally interpretable. Appropriate laboratory instruments are selected based on the quantity being measured; for example, balances are used to measure mass in grams, and graduated cylinders are used to measure liquid volume in milliliters.

The metric system is built upon a set of base units commonly used in biology. The meter (m) is the standard unit for measuring length; the liter (L) is used to measure volume, and the gram (g) is used to measure mass. Temperature in biological laboratories is typically measured in degrees Celsius (°C). All recorded measurements must include both a numerical value and the proper unit. For example, a measurement of 2 m indicates a length that is twice the base unit of length, the meter. Standard unit symbols and abbreviations are allowed and should be used consistently when recording data.

A primary advantage of the metric system is the ease with which units can be converted. Because the system is based on powers of ten, conversions involve multiplication or division by 10 or multiples of 10. For example, one decameter is equal to ten meters, and one kilometer is equal to one thousand meters. This base ten structure simplifies calculations, reduces the likelihood of errors, and makes the metric system especially well-suited for scientific measurements and laboratory applications.

Base Units of Measure

- Length: meter (m)
- Mass: gram (g)
- Volume: liter (L)
- Time: seconds (s)
- Temperature: degree Celsius ($^{\circ}\text{C}$)

Table 1.1 Metric Prefixes

Prefix	Unit	Multiplier	Scientific Notation
Kilo-	k	1,000	10^3
Hecto-	h	100	10^2
Deca-	da	10	10^1
One	base (m, L, g)	1	10
Deci-	d	$0.1 = 1/10$	10^{-1}
Centi-	c	$0.01 = 1/100$	10^{-2}
Milli-	m	$0.001 = 1/1,000$	10^{-3}
Micro-	μ	$0.000001 = 1/1,000,000$	10^{-6}

Table 1.2 Mnemonics to help memorize the metric prefixes

Mnemonic Prefix	King Kilo-	Henry Hecto-	Does Deca-	Usually Base units	Drink Deci-	Chocolate Centi-	Milk Milli-	Mondays Micro
unit	k	h	da	m, L, g	d	c	m	μ

In Metric conversions, when you are converting a smaller unit to a larger unit, you move the decimal point to the left for the proper number of steps. Keep in mind each time you move the decimal point you are dividing by 10. When you are converting from a larger unit to a smaller, you will move the decimal point to the right. This means each time you move the decimal point you are multiplying by 10.

Steps for Converting Metric Units

1. Write down the number you are converting, for example (100 cm). Then enter the decimal point. It is always right after the ones place to the right of the number.

100 = 100.

2. If you want to convert 100 cm to meters (m) you will now look at your chart and determine how many “steps” you must move the decimal to the right or left. From centimeter to meters, you must take 2 steps to the left. That means you must move your decimal two places to the left.

3. **100. cm = 1.00 meters**

Lab Activity .11: Metric Conversion Practice

Using the steps above, complete the following problems:

1. 50 mm = _____ cm
2. 50 cm = _____ km
3. 700 mL = _____ L
4. 30 m = _____ μm
5. 3 dm = _____ m
6. 15 kg = _____ cg
7. 55 L = _____ mL
8. 52 mg = _____ μg

Lab Activity 1.2: Measuring Using the Metric System

i. Taking Linear Measurements with a Ruler

Figure 1.1: Metric ruler displays measurements alongside imperial units.

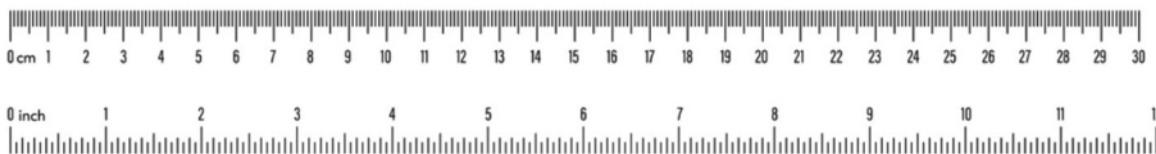


Image of a metric ruler displaying centimeter and millimeter markings, with numbered intervals and smaller subdivisions used to measure length accurately.

Figure 1. Example of a metric ruler. By Russetburbank - Own work, CC BY-SA 4.0.

The fundamental unit of length in the metric system is the meter (m). In this laboratory exercise, length measurements will be made using centimeter (cm) rulers. As illustrated on a meter stick or metric ruler, one meter is equal to 100 centimeters (cm). When examining a metric ruler, each centimeter is subdivided into smaller intervals. Between each numbered centimeter marking, there are ten equally spaced hatch marks, each being one millimeter (mm).

Therefore, ten millimeters are equal to one centimeter. Familiarity with these scale divisions is essential for accurately measuring and recording length in biological laboratory experiments.

Materials

- Rulers
 - 4-5 objects of various sizes

Procedure

1. Using a centimeter ruler, record the length, width, and height, in centimeters or mm, as directed, of each object provided to you by your instructor.
2. Record your results in **Table 3**.

Results

Record the measurements in the Table 1.3 below:

Table 1.3 Linear Measurements and Unit Conversions

Object	Length (cm)	Width (cm)	Height (mm)
Black metal object			
Storage bins			
Test tube rack			
Your Lab Bench			
Glovebox			

Lab Activity 1.3: Taking Mass Measurements with an Electronic Balance

Weight measurements in science are also expressed in metric units. The basic unit is the gram (g). The electronic balances you will be using today are gram balances. The model you will be using will accurately measure 0.01 grams. There are 1000 grams in a kilogram. One of the most common units used in labs is the milligram (mg). There are 1000 milligrams in a gram. If you need a very small amount of something, you measure it in micrograms (μg). There are 10^6 μg in a gram. Some conversions are shown below:

$$1000 \text{ g} = 1 \text{ kg}$$

$$1 \text{ g} = 1000 \text{ mg}$$

$$1 \text{ g} = 1,000,000 \text{ } \mu\text{g} \text{ (} 10^6 \text{ } \mu\text{g)}$$

$$1 \text{ mg} = 1000 \text{ } \mu\text{g} \text{ (} 10^3 \text{ } \mu\text{g)}$$

Materials

- Gram balance
- Objects used in the earlier activity

Procedure

1. Press the on button and wait for the balance to display zeros on screen.
2. If the screen does not display zeros, press the “zero” or “tare” button.
3. Once the machine displays zeros (0.00 g), place your object on the center of the platform.
4. Wait for the scale to achieve a stable reading (numbers are not fluctuating).
5. Record your mass in grams in Table 1.4 for each object starting from smallest to largest.

Results

Record your results in table 1.4 as grams (g) and then convert those masses to kg and mgs.

Table 1.4. Weight Measurements and Unit Conversions

Object	Mass (g)	Mass (kg)	Mass (mg)
Black metal object			
Test tube			
250 ml beaker			
Weigh boat			
your cell phone			

Lab Activity 1.4: Volumetric Measurements

The metric unit for volume is the **Liter (L)**. There are 1000 milliliters (mL) in one liter. Another common unit in volume is the microliter (μL). There are 10^6 μL in one liter and 1000 μL in one milliliter. Some common conversions are shown below:

$$1 \text{ L} = 1000 \text{ mL}$$

$$1 \text{ L} = 1,000,000 \text{ } \mu\text{L} \text{ (} 10^6 \text{ } \mu\text{L)}$$

$$1 \text{ mL} = 1000 \text{ } \mu\text{L} \text{ (} 10^3 \text{ } \mu\text{L)}$$

You will need to become familiar with the diverse types of instrumentation and glassware that you will be using throughout this semester. Today, we will focus on glassware and devices that measure larger volumes of liquid. You will also decide when a particular device is appropriate to use, based on the volume that you are dispensing. The types of measuring devices are quite different if you want to measure and dispense a liter versus a milliliter!

Graduated Cylinder



Image showing four graduated cylinders of different capacities (10 mL, 25 mL, 50 mL, and 100 mL), each with measurement markings and narrow cylindrical shapes used for measuring liquid volumes.

Figure 1.2. Several types of graduated cylinders: 10ml, 25ml, 50ml, and 100ml. Praphai Donphaimueang - Own work, 15 March 2016, CC BY-SA 4.0.

You will use this to dispense large volumes that are more than 10 mL. You will be using various size graduated cylinders, ranging from 10mL – 2000 mL (2L), in this lab. The volume of fluid in a graduated cylinder is measured by reading the bottom of the **meniscus** or the curved upper surface of the fluid in the graduated cylinder (red arrow)

Lab Activity 1.5: Serological Pipettes:

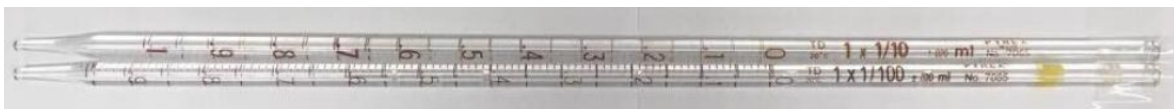


Figure 1.3. Examples of Serological Pipets, By Paweena.S - Own work, CC BY-SA 4.0. Image showing examples of serological pipettes, long transparent tubes with graduated markings,

designed for measuring and transferring specific volumes of liquid using a pipette pump.

These pipettes accurately dispense volumes of 1mL to 10mL and can be used for volumes up to 50 mL. You will be using mostly 1mL and 10mL serological pipettes in this class.

Reading a Meniscus

When measuring liquid volume in a pipette or graduated cylinder, it is important to read the **meniscus**, the curved surface of the liquid. Most liquids (like water) form a **concave meniscus**, meaning the surface curves downward.

To obtain a correct measurement:

- Position your eyes **level with the liquid surface** to avoid parallax errors.
- Read the measurement at the **lowest point of the meniscus** (the bottom of the curve) for those liquids that form a concave meniscus. For liquids that form a **convex meniscus** (such as mercury), the measurement should be read at the **top of the curve**.
- Ensure the instrument is **upright on a flat surface**, and the liquid is still.

Careful alignment and consistent technique are essential for precise volume measurements.

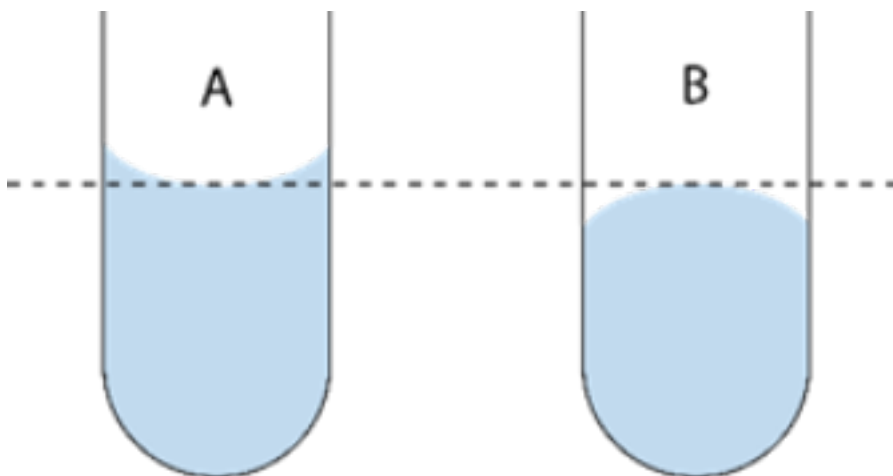


Figure 1.4: Demonstration on how to read a meniscus properly. By Reading_the_meniscus.png: User:Jleedevderivative work: Rehua (talk) - This file was derived from: Reading the meniscus.png: CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=18062727>

Materials

- 100 ml beaker
- Distilled H₂O
- 50 ml graduated cylinder

- Gram scale
- 1- and 10-ml serological pipettes
- 1 large and 1 small pipette pump
- 1mL microcentrifuge tubes

Procedure

Graduated Cylinder Measurements

1. Obtain a 100 mL beaker. This is the container you will use to weigh your water. It is not what you will use to measure in this experiment.
2. Place the beaker on the balance and tare it.
3. The **target** amount of water you will be measuring using a graduated cylinder is 42 mL. This has been recorded in Table 5.
4. Pour 42 mL of water into a 50 mL graduated cylinder. Be sure to read from the bottom of the meniscus.
5. Pour the 42 mL from the graduated cylinder into the tared beaker.
6. Determine the weight of the water and record as “**Weight of Water.**”
7. Convert this weight to mL. Water has a density of 1g/mL, therefore the number of g= number of mL (e.g. 50 g = 50 mL). Record this number as “**Actual Volume Dispensed.**”
8. Calculate the % **error** for each of your measurements as follows:

$$\text{Percentage Error} = \frac{|\text{Measured Value} - \text{True Value}|}{\text{True Value}} \times 100$$

9. If you are not within an error range of +/- 5% then try again.
10. **Note:** Percent error is always reported as an absolute value, so no negative sign is included. For example, a calculated value of -5% is reported as **5% error**.

Lab Activity 1.6: Serological Pipette Measurements

1. Dry the 100 mL beaker you used previously, with a dry paper towel. Tare it.
2. Get a **10 mL serological pipette** and its pipette pump. Insert the blunt end of the pipette into the pipette pump. Use the dial to draw 6.6 mL of deionized water into the pipet from a beaker filled with water.
3. Dispense the water into the tared beaker, dialing the pump in the opposite direction. To get the last bit out of the pipet, quickly dial one way then the other.

4. Determine the weight of the water in the beaker. If you accurately dispensed 6.6 mL; the weight difference should be close to 6.6 g. Record, it as “**Weight of Water.**”
5. Convert the weight to mL and record as “**Actual Volume Dispensed**”
6. Calculate the % error as described in the earlier section and record on Table 1.5.
7. If you are not within an error range of +/- 5% then try again
8. Repeat steps 2-8 with a **1 mL serological pipette** to dispense 0.6 mL of water into a tared microcentrifuge tube.

Results

Table 1.5 Comparison of Target, Actual Volumes and Percent Error Dispensed Using Different Measurement Devices

Device used	At-tempt	Target volume dispensed (mL)	Weight of Water (g)	Actual Volume Dispensed (mL)	% Error
50 mL graduated cylinder	trial #1	42			
50 mL graduated cylinder	trial #2	42			
10 mL pipet	trial #1	6.6			
10 mL pipet	trial #2	6.6			
1 mL pipet	trial #1	0.6			
1 mL pipet	trial #2	0.6			

Lab Activity 1.7: Accuracy and Precision

When making measurements, both precision and accuracy are extremely important. **Precision** of a measurement refers to the **closeness of repeated measurements or the reproducibility** of the measurement. Precision can be affected by the measuring instrument. If a student uses a thermometer calibrated to the nearest degree to measure the temperature of a beaker of water and a second student uses a thermometer that is calibrated to the nearest ten degrees for the same exercise, the reproducibility or precision of measurements will be greater for the student using the thermometer calibrated to the nearest degree. The thermometer with the greatest number of calibrations (smallest increments) is the most precise, and thus the

reproducibility is greater. Accuracy, described as how close a measurement is to its true value, however, is dependent upon both the **precision of the measuring instrument** and the **technical skills** of the individual taking the measurement. You will perform an experiment to determine which instrument is most accurate and precise for measuring 10ml of liquid.

Materials

- 100 ml beaker
- Distilled Water
- 50 ml graduated cylinder
- Gram scale
- 10 ml serological pipette
- Pipette pumps

Procedure

1. Obtain a 100 mL beaker. Place the beaker on the scale and hit “tare.” This will cancel the weight of the beaker.
2. Obtain a 50 mL beaker and use it to measure 10 mL of deionized water. Pour the measured 10 mL of water into the 100 mL beaker on the scale. Record the weight in grams in the Table below.
3. Convert this weight to mL. Record this number as “**Actual volume dispensed.**”
4. Empty the beaker on the scale and dry it well with a dry paper towel. Place it back on the scale and tare.
5. Obtain a 50 mL graduated cylinder and measure 10 mL of deionized water. Pour the measured 10 mL of water into the 50 mL beaker on the scale. Record the weight (g) and volume (ml).
6. Obtain a 10 mL serological pipet and measure 10 mL of deionized water. Pour the measured 10 mL of water into the 50 mL beaker on the scale. Record the weight (g) and volume (ml).

Results

Table 1.6: Volume, Weight, and Actual Volume Results:

Measuring Device	Volume Dispensed (ml)	Weight of Water (g)	Actual Volume Dispensed (ml)
100 ml beaker			
50 ml graduated cylinder			
10 ml serological pipet			

Conclusion

Use your data and the questions below to write your conclusion below:

1. For the volume measurements, which glassware used to measure 10 mL was the most precise? Why?
2. Which measuring device was the least precise?
3. For the glassware used to measure 10 mL, which instrument was the most accurate?
4. Was the instrument that was most precise, the same one that is the most accurate? Why or why not?

Lab Activity 1.8: Graphing on an Excel Spreadsheet

1. Open a blank MS Excel file. Save it using the **File** tab “**Save as**” choice. Name it, “Lab 1 Graph_ your last name”. Make sure you save it in a location on the computer hard drive that you remember.
2. Pick any cell on any column and type in **Amount of Water (ml)**. On the next cell to the right type in **Weight (g)**.
3. Under the **Amount of water** column type in the volumes of water, whose weights were measured during the experiment recorded on Table 1.6 (independent variable). Under the **Weight** column, type in the corresponding weights measured. (see figure on top right)
4. Click on the **Insert** Tab on top of the page (see Figure 1.5)

5. Select all the cells with text or numbers in them (Press **Shift** and select all the cells while holding the shift button). With the cells selected click on **Recommended Charts** in the menu items (See Figure 1.5)

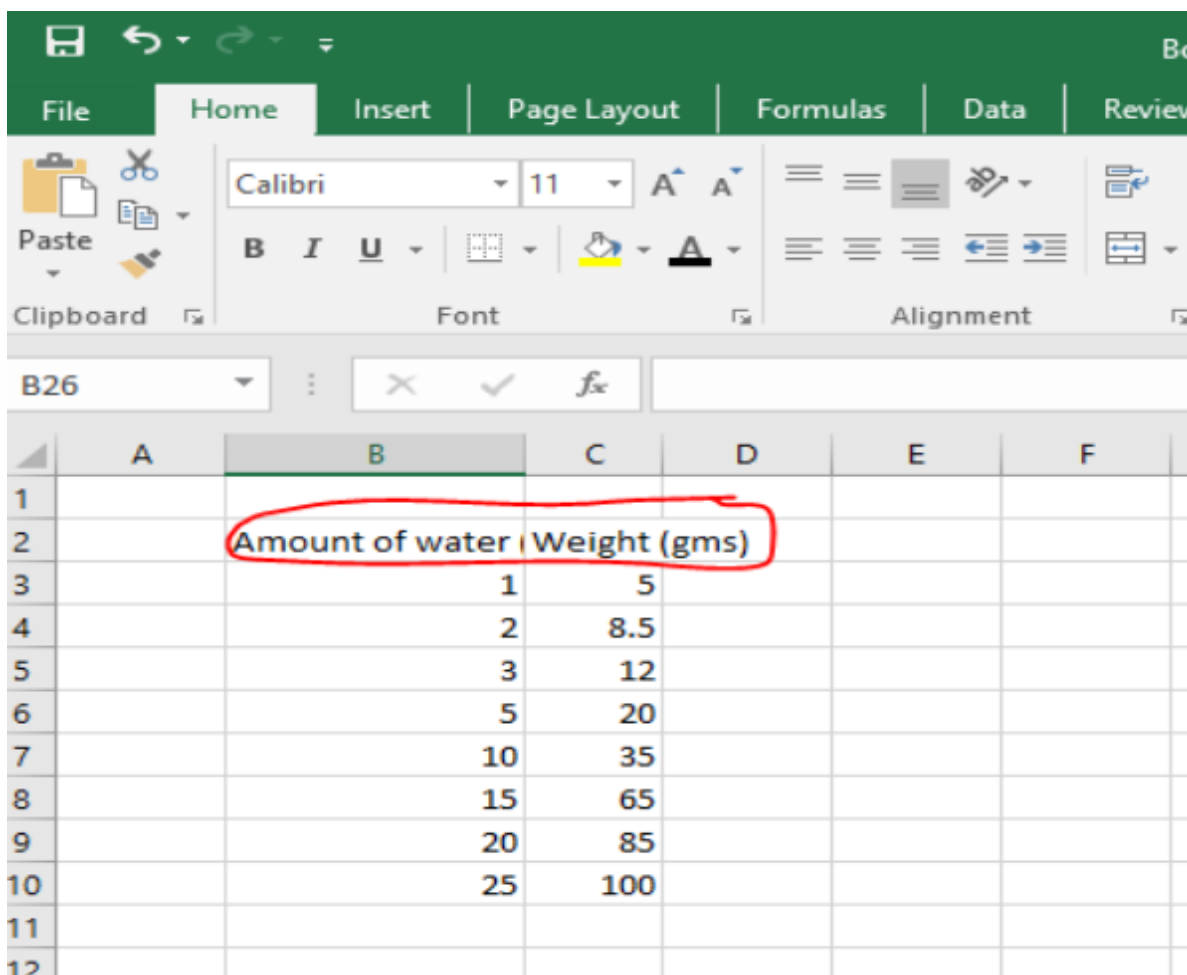
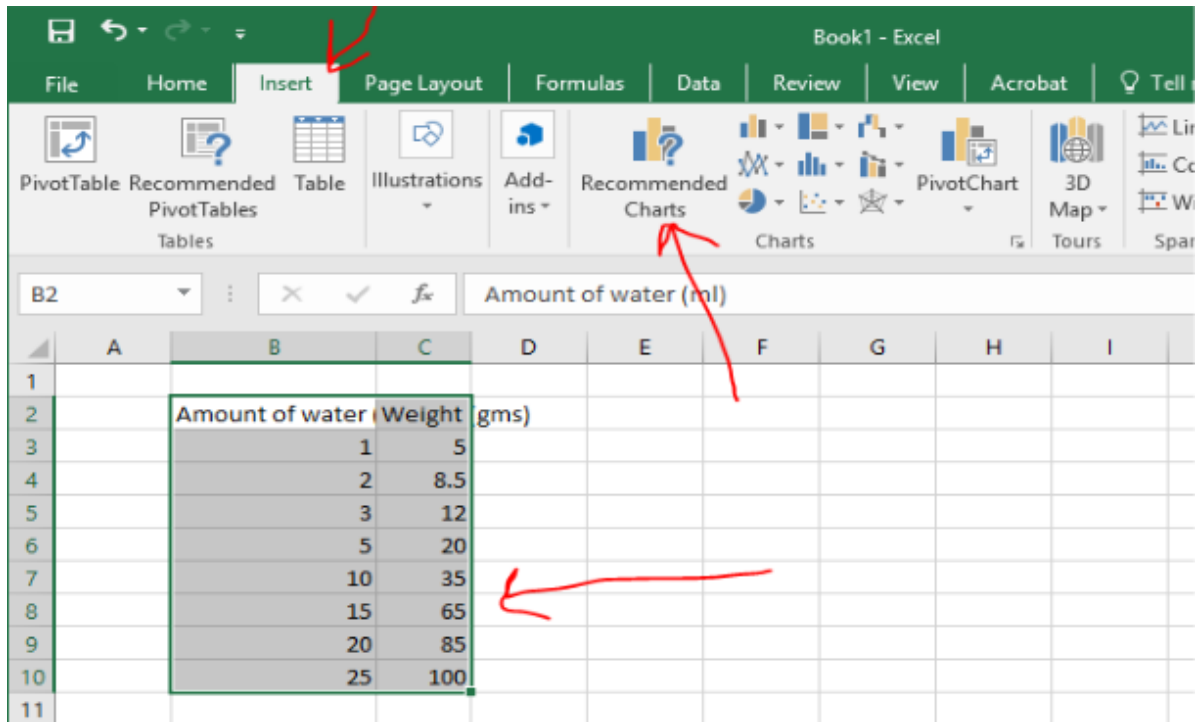


Figure 1.5: A screenshot showing data being entered into an Excel spreadsheet, with columns recording the volume and weight of water measured in a laboratory container.

6. A small window opens with two tabs on the top. Click the first one on the right that says **All Charts** and pick **XY Scatter** and hit OK
7. The window will close, and a scatter plot graph will appear on the main Excel sheet (Fig. 1.7).
8. Click inside the graph box and three small open shows up to the right of the graph box.
9. Click on the green + sign and check on the boxes next to **Axis**, **Chart title**, **Axis titles**, and **Gridlines**. The rest of the boxes can stay unchecked.

10. Rename the Axis tiles with **Amount of Water** (X-axis) and **Weight** (Y-axis). Rename the Chart tile to say, “**Volume-Weight Relationship.**”
11. Save the file (**Control + S**).



screenshot of a computer

AI-generated content may be incorrect.”}

Figure 1.6: A screenshot showing multiple cells selected in an Excel spreadsheet, containing text and numbers. The top menu highlights the “Recommended Charts” choice, showing the step used to create a chart from the selected data.

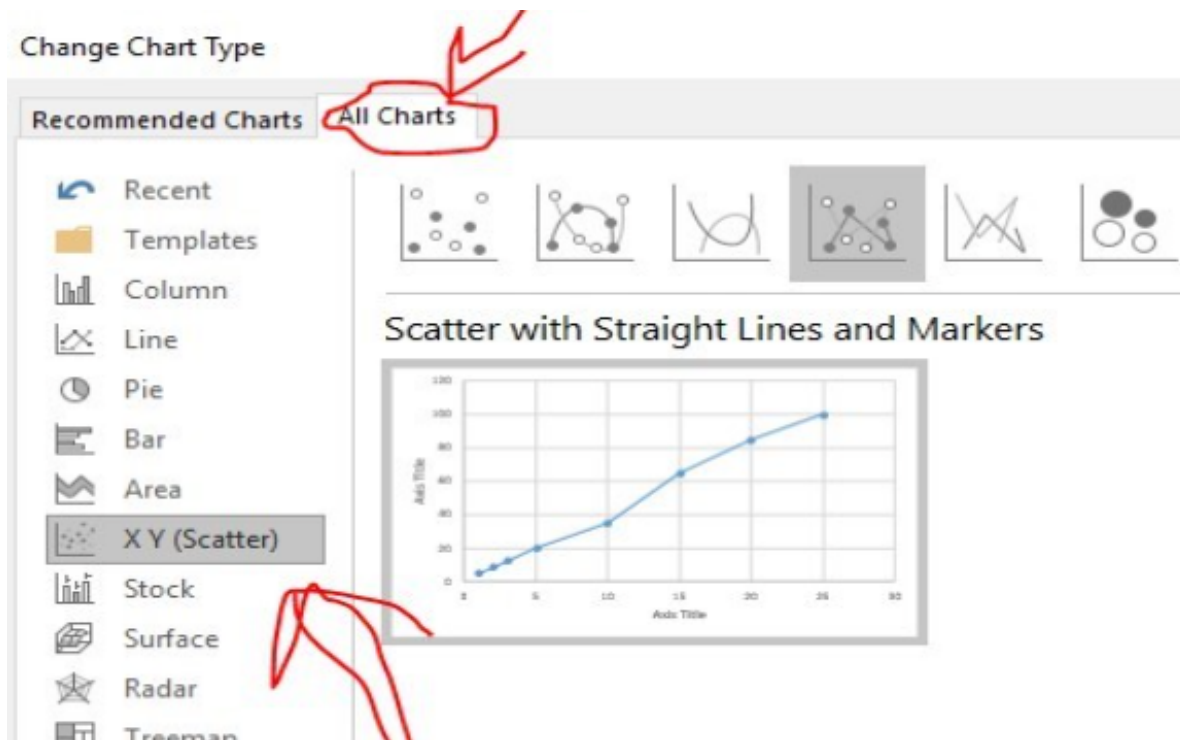


Figure 1.7: A screenshot of an Excel chart window showing two tabs at the top. The “All Charts” tab is selected, and the XY (Scatter) chart option is chosen, with an OK button highlighted to create the chart.

Post Lab Activity 1.1: Questions

Students must complete these questions after completion of the lab and give them to your instructor for review

Multiple choice

1. What should you do if a chemical gets in your eyes?
 - A. Ignore it
 - B. Flush with water immediately
 - C. Leave lab
 - D. Tell a classmate
2. Where should broken glass be disposed of?
 - A. Trash
 - B. Sink
 - C. Designated container
 - D. Floor

3. Why should you not eat in the lab?
 - A. Messy
 - B. Contamination risk
 - C. Rules only
 - D. No reason

Short answer

4. Describe one hazard you observed or learned about.
5. How does PPE reduce risk in the lab?
6. Which measuring device was most precise for measuring liquid volume, and why?
7. Which device was the least precise for measuring 10 mL of liquid?
8. Was the most precise instrument also the most accurate? Explain your answer using your data.
9. Why is it important to read the bottom of the meniscus when measuring liquid volume?
10. How does the base-10 structure of the metric system reduce calculation errors?
11. What source of error could affect mass or volume measurements in this lab?
12. Explain why SI units are essential for communicating scientific data.
13. Convert 250 mL to liters.
14. A cell is 5 μm wide. How many meters is this?
15. Explain one advantage of using Celsius in biology experiments.
16. Which metric prefix would be most appropriate for measuring bacterial size, and why?
17. Describe a potential laboratory error that could occur if imperial units were used instead of SI units.
18. Describe one hazard you observed or learned about.
19. How does PPE reduce risk in the lab?

Additional Student Resources

Metric Conversions and Scientific measures. From New York City College of Technology, CC BY NC SA.

<https://openlab.citytech.cuny.edu/bio-oer/biology-basics/units-of-measure/>.

Quantitative Skills.

This site has instructions on variables, graphing a line, slope activity, and creating a line of best fit. From New York City College of Technology, CC BY NC SA.

<https://openlab.citytech.cuny.edu/bio-oer/biology-basics/quantitative-skills/>.

Reporting Science

This is a description of how to write lab reports. From New York City College of Technology, CC BY NC SA.

<https://openlab.citytech.cuny.edu/bio-oer/biology-basics/reporting-on-science/>.

1.11 Appendix: Laboratory Safety and Measurement Terms

Term	Definition
Accuracy	How close a measured value is to the true or accepted value.
Balance	An instrument used to measure mass.
Base Units	Standard units in the metric system such as meter, gram, liter, and second.
Biological Hazard	Living organisms or biological materials that may cause harm.
Chemical Hazard	A harmful substance that can cause injury or damage.
Contamination	The introduction of unwanted substances can affect results.
Conversion	The process of changing a measurement from one unit to another.
Data Recording	The accurate documentation of observations and measurements is important.
Density	The mass of a substance per unit of volume is incorrect.
Dependent Variable	The variable that is measured or observed in an experiment.
Experimental Error	Deviation from the expected result due to limitations or mistakes.
Eyewash Station	A safety device used to flush the eyes in case of chemical exposure.
Graduated Cylinder	A piece of laboratory glassware used to accurately measure liquid volume.
Graphing	The visual representation of data for analysis is important.
Hazard	A potential source of harm such as chemicals, heat, or sharp objects.
Independent Variable	The variable that is changed or controlled in an experiment is correct.
Laboratory Procedure	Step-by-step instructions followed during an experiment.
Laboratory Safety	Practices are used to prevent accidents and ensure safe working conditions in a lab.
Length	The measurement of distance from one point to another is important.
Mass	The amount of matter in an object.
Meniscus	The curved surface of a liquid in a container used for accurate measurement.
Metric Prefix	A factor that indicates multiples or fractions of base units (e.g., kilo-, milli-, micro-).

Term	Definition
Metric System (SI)	A decimal-based system of measurement used in science worldwide.
Percent Error	A calculation that describes how far a measured value is from the true value, expressed as a percentage.
Personal Protective Equipment (PPE)	Protective clothing and gear like goggles and gloves used to prevent injury.
Physical Hazard	Environmental factors such as heat, glass, or equipment that can cause injury.
Pipette	A tool used to measure and transfer small volumes of liquid accurately.
Precision	The consistency or repeatability of measurements is important.
Reproducibility	The ability to obtain consistent results across repeated trials is important.
Scatter Plot	A graph used to display relationships between two variables.
Scientific Measurement	The process of obtaining quantitative data using standardized methods.
Tare	The process of resetting a balance to zero before measuring. ⁸
Temperature	A measure of the amount of heat energy in a substance.
Volume	The amount of space an object or substance occupies.
Waste Disposal	The proper handling and discarding of laboratory materials.

2 Use of the Microscope

2.1 Introduction: Use of the Microscope

In this lab, you will learn how to use one of the most important tools in biology: the **compound light microscope**. Microscopes allow scientists to observe objects that are too small to be seen with the naked eye, such as cells, tissues, and microorganisms. Learning how to effectively use a microscope is a fundamental skill that you will use throughout this course and in many scientific careers.

During this lab, you will become familiar with the **parts of the microscope** and how each part works together to produce a clear image. You will focus on specimens using different magnifications, starting with low power, and working your way up to higher magnifications. As you increase magnification, you will notice critical changes in what you see, including how much of the specimen is visible (**field of view**) and how much of it stays in focus (**depth of field**).

You will also learn how to prepare your own slides using a technique called a **wet mount**, which allows you to see living or freshly prepared specimens. In addition, you will practice making careful observations and recording what you see through drawings and notes—an essential part of scientific investigation.

Proper handling and care of the microscope will be emphasized throughout the lab. Using the microscope correctly not only protects delicate equipment but also ensures that your observations are accurate and reliable.

By the end of this lab, you will have the skills needed to confidently use a microscope to explore the microscopic world and better understand the structure of living organisms.

2.2 Objectives

By the end of this laboratory exercise, students will be able to:

1. Correctly identify and use the scanning, low power, and high-power objective lenses of a compound light microscope.
2. Define and compare field of view and depth at different magnifications.

3. Demonstrate proper preparation of a wet mount using cultured organisms or pond water.
4. Use fine focus to determine the relative vertical position of objects within a specimen.
5. Observe and accurately document microscopic specimens at increasing magnifications.
6. Explain how magnification affects image clarity, field of view, and depth of field.
7. Apply proper microscope handling, cleaning, and storage procedures.

2.3 General Rules:

- Always carry the microscope using two hands. The easiest way to do this is to place one hand holding the arm while the other hand is supporting the base.
- Do not touch the objective lenses. The lenses should originally be set in the scan position and when stored, the microscope scanning lens should be set in place.
- If needed, the ocular and objective lenses must be cleaned with a special lens paper and lens cleaning solution.
- Report any malfunctions of the microscope to your instructor; do not try to repair the scope yourself.

2.4 Returning the Microscope to the Cabinet.

- Lower the stage.
- Rotate the turret so that the scanning (4x) lens is in position over the stage.
- Remove the lens from the stage and clean the lens before replacing it in its storage box.
- Secure the cord by wrapping it around the correct holder on the back of the microscope.
- Replace the dust cover.
- Carry the microscope back to the storage cabinet with **two hands!**

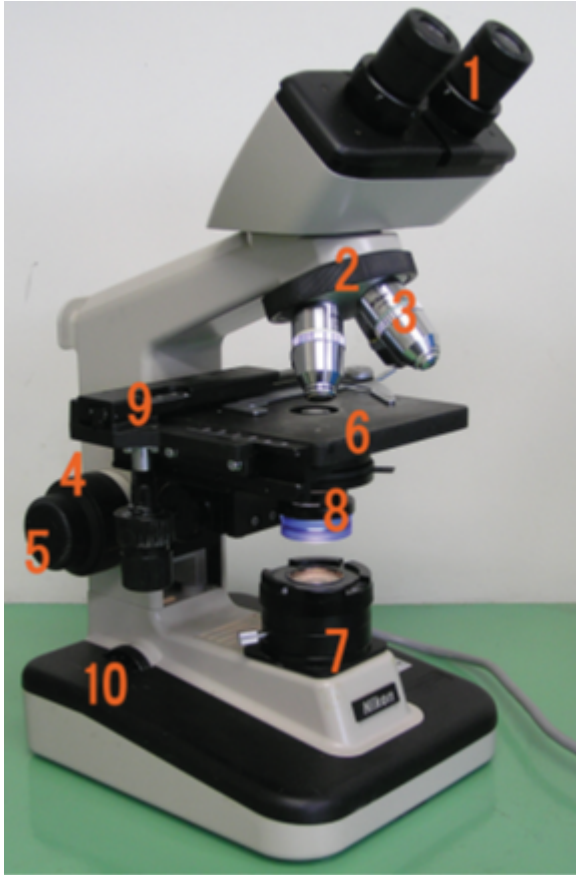
2.5 Prelab Activities

2.6 Introduction

These pre-laboratory activities are designed to familiarize students with the structure and function of the compound microscope. Students will identify key microscope components, match each part with its function, and apply their understanding to written responses. Completion of these activities prior to the lab session will help ensure proper microscope use and improve observational accuracy during laboratory work.

2.7 Prelab Activity 2.1. Terms to define:

- Arm
- Neck
- Body
- Base
- Ocular lens
- Objective lens
- Rotating nosepiece or Turret
- Course and Fine focus knobs
- Mechanical stage
- Iris diaphragm
- Condenser adjustment knob



Enumerated diagram of a typical bright-field microscope showing major components, including the ocular lens, head, revolving nosepiece, objective lenses, stage, condenser, iris diaphragm, light source, coarse and fine focus knobs, arm, and base.

Figure 2.1 Components of a Typical Bright Field Microscope.

2.8 Prelab Activity 2.2

Identify the following parts and functions of the microscope above.

1. _____
2. _____
3. _____
4. _____
5. _____

6. _____
7. _____
8. _____
9. _____
10. _____

2.9 Prelab Activity 2.2

Match the microscope part with its definition.

Microscope Part

- _____ Condenser
- _____ Revolving Turret
- _____ Mechanical stage
- _____ Ocular lens
- _____ Power switch
- _____ Iris diaphragm
- _____ Arm
- _____ Depth of field
- _____ Fine focus
- _____ X-Y stage controls
- _____ Light source
- _____ Total magnification
- _____ Head
- _____ Base
- _____ Objective lens
- _____ Course focus
- _____ Field of view

2.10 Definitions

- (a) The light source at the bottom of the microscope, also known as the illuminator.
- (b) Also known as the body- carries the oculars.
- (c) The microscope supports.
- (d) The part that connects the head and the base.
- (e) The lens closest to the eye
- (f) The major lenses closest to the specimen
- (g) Also known as the nosepiece-holds the objective lenses
- (h) Knob that makes large scales adjustments in the focus
- (i) The knob that makes small adjustments in the mechanical stage.
- (j) Set of lenses that collect and focus the light onto the specimen.
- (k) Controls the amount of light reaching the specimen.
- (l) Controls that move the slide vertically and horizontally on the stage
- (m) Calculated by multiplying the ocular magnification by the objective magnification.
- (n) The visible area one can see through the oculars.
- (o) The measure of the thickness or the specimen in focus
- (p) The switch used to turn the microscope on and off.
- (q) The section in which the specimen is placed for viewing.

2.11 Prelab activity 2.3

Answer the following questions in a brief paragraph.

- What is the field of view, and how does it change as magnification increases?
- Define depth of field in microscopy.
- Which objective lens provides the greatest depth of field? Why?
- Why is it important to center a specimen before switching to a higher magnification?
- What is a wet mount, and what type of specimens are commonly observed using this technique?

- What is the purpose of using the fine adjustment knob instead of the coarse adjustment knob at high magnification?

2.12 Lab Activities

2.13 Introduction: Magnification and Microscope Use

This laboratory activity introduces students to the proper use of the compound light microscope, a fundamental tool in biological sciences. Students will learn to identify microscope components, understand how magnification is achieved, and develop skills in focusing on specimens at increasing levels of magnification. Emphasis will be placed on proper handling, accurate observation, and the relationship between magnification, field of view, and depth of field. Mastery of these techniques is essential for examining biological specimens and conducting accurate scientific observations.

2.14 Magnification

Student scopes used in our classes have four objective lenses (i.e., those that are closest to the specimen); scanning, low power, high power, and oil immersion. The set of lenses closest to the eye are the ocular lenses.

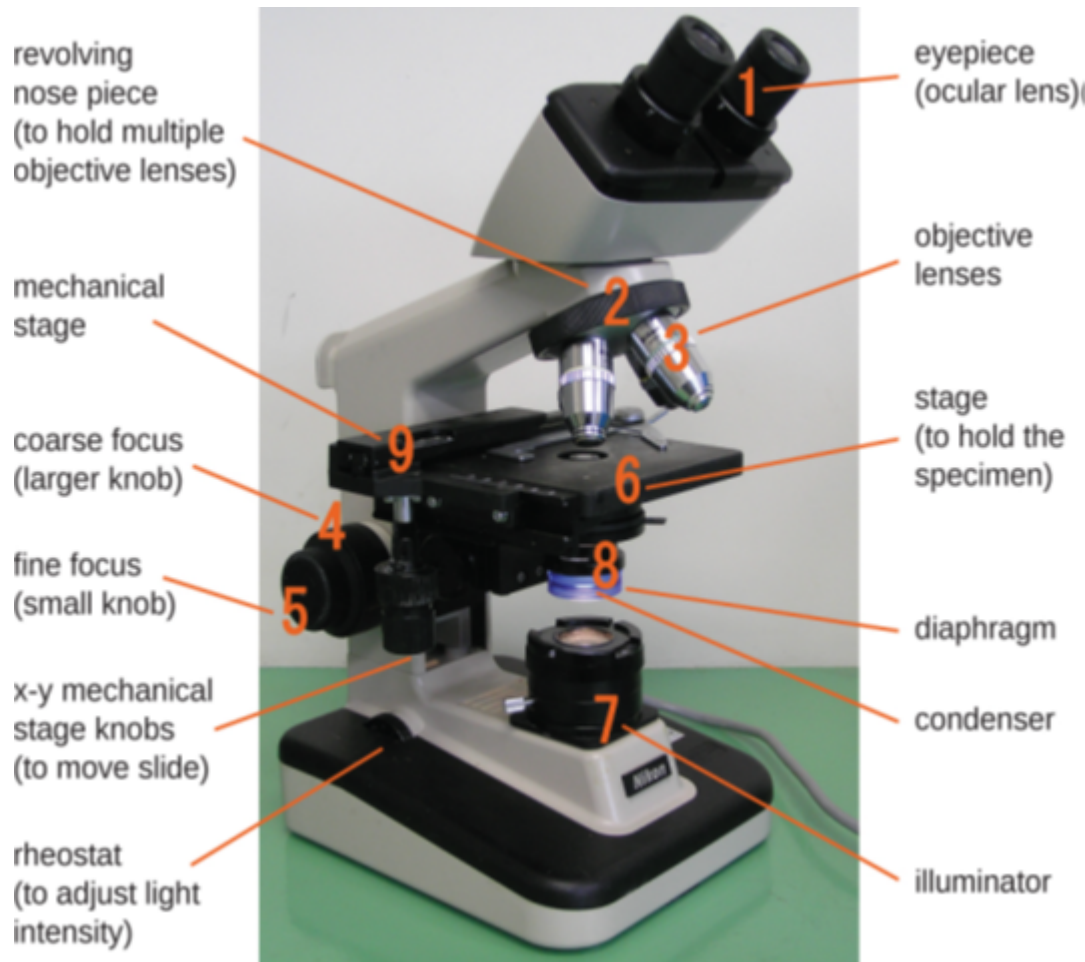
General Procedures

Microscope components

2.15 Lab Activity 2.1

- Carry the microscope with **two hands** and bring it to your lab bench
Clear your workspace for safe and comfortable use
- Collect slides:
 - Metric ruler
 - Letter “e”
 - Crossed colored threads
 - Prepared organism slides
- Clean ocular and objective lenses with lens paper
Set the nosepiece to the **scanning (lowest power) lens**

- Plug in the microscope
- Identify and label microscope parts



Labeled diagram of a bright-field microscope showing key components, including the ocular lenses, head, revolving nosepiece, objective lenses, stage, mechanical stage controls, condenser, iris diaphragm, light source, coarse and fine focus knobs, arm, and base.

Figure 2.2 Components of a Typical Bright Field Microscope, labelled.

Key: Components of a typical bright field microscope.

1. Ocular lens
2. Head
3. Revolving nose piece (turret)
4. Objective lens
5. Arm

6. Mechanical stage
7. Stage clips
8. Diaphragm
9. Condenser
10. Fine focus
11. Course focus
12. X-Y Mechanical stage knobs
13. Light source (Illuminator)
14. Rheostat
15. Illuminator switch

2.16 Lab Activity 2.1

Record the magnifications of the lenses on your microscope in the table below.

Table 2.1. Ocular and Objective Lens Magnifications and Total Magnification

Lenses	Ocular Lens	Objective lens	Total Magnification
Scanning power: dry			
Red lines			
Low power: dry			
Blue lines			
High Power: dry			
Yellow lines			
Oil Immersion: oil			
Black/white lines			

To figure out the total magnification simply multiply the ocular lens magnification by the objective lens magnification. For example, if the Ocular lens magnification is 10X and the scanning lens magnification is 4X, then the total magnification will be 40X. Complete the table above.

2.17 Focusing on Specimens

For biology 110 labs you will only be focusing on a total magnification of 400x. You will have to use the oil immersion lenses (1000x) in your Microbiology (230) classes. For your first practical, you will have to focus clearly on a specimen at a total magnification of 400x. You will have two minutes to complete this task.

Focusing on a specimen may be broken down into a three-step process.

1. Scanning Lens (4x):

- Begin focusing on a specimen using the Scanning lens. First, adjust the mechanical stage so that it is at its lowest position. You can do this by turning the course focus knobs away from your body.
- Turn on the light. And adjust the ocular lenses so that your field of view is a circle of light.
- Place and center the specimen on the mechanical stage.
- Look into the oculars and slowly raise the mechanical stage by turning the course focus knob towards your body. Stop when it comes to focus. You may adjust the focus using the fine focus knob that is needed.
- Center the specimen. Adjust the X-Y stage knobs so that your specimen is in the center of your field of view. This position will enable you to see the specimen when you adjust the lens to the next higher magnification. If the specimen is found to the side, then you may lose the specimen from your field of view when you adjust your objective lens to the next higher magnification.

2. Low Power (10x)

- Next, while looking into your objective lenses, turn the turret clockwise (to the right) to 10x power.
- Adjust the focus. Do not use the course focus knob to adjust your focus. Only use the fine focus knob at this point. If you use the course focus knob, then you may adjust your focus too far out of focus and may have to start at the Scanning power again.
- Center your specimen

3. High Power (40X)

- Finally, switch to the high-power lens (40X) by turning the lens to the right (clockwise). At this point only use the fine focus to adjust the focus.
- If the color of light is too yellow, you may adjust the rheostat to change the color of light to a white light.
- If the light is too dark, you may adjust the diaphragm.

2.18 Measuring and Calculating the Field of View

The field of view is the area of the specimen you can see through the ocular lenses. When you adjust your magnification, the diameter will decrease at higher magnifications. The image below, figure 2.3, is an example of a field of view.

How to find the field of view:

- Obtain a prepared slide with a ruler and place it on the stage.
- Adjust the ruler so that the edge of the ruler is visible as a series of tic marks/lines that lie across the diameter or the field of view as in the figure below.

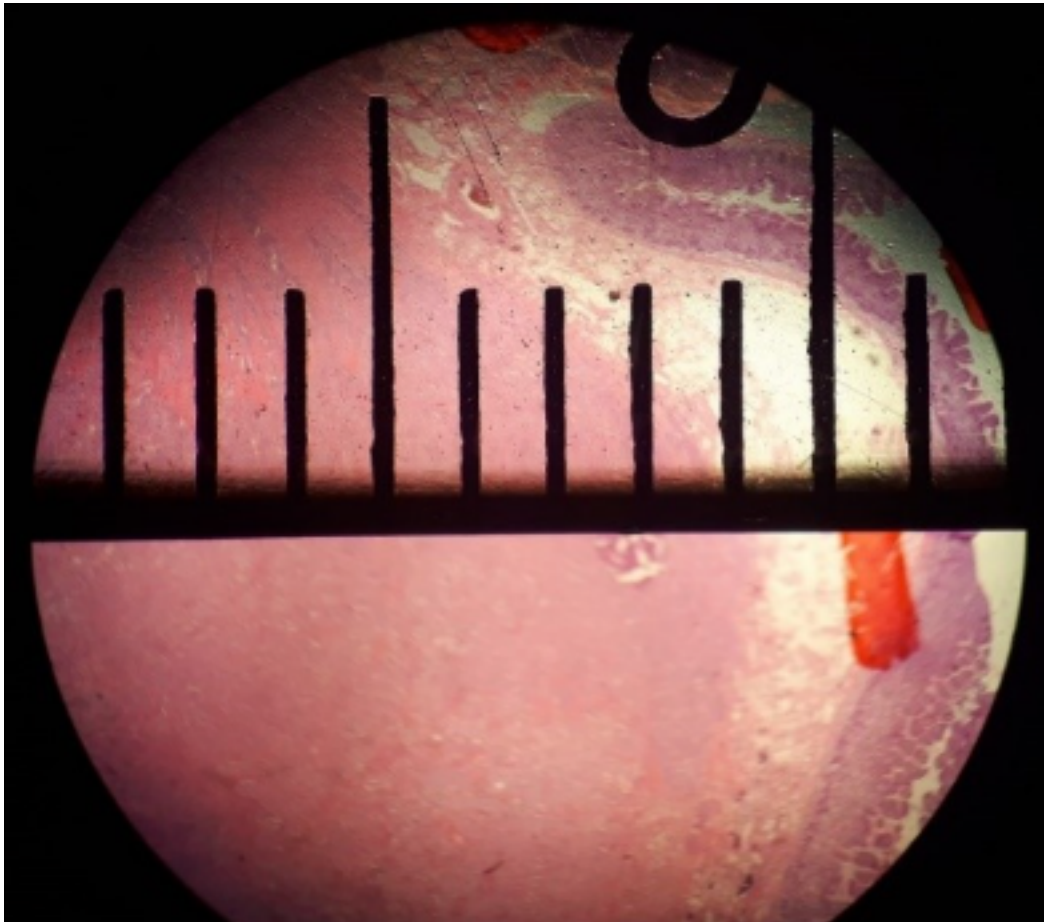


Figure 2.1: https://upload.wikimedia.org/wikipedia/commons/thumb/6/69/Field_of_view_diameter_in_microscopy.jpg/1024px-Field_of_view_diameter_in_microscopy.jpg

Microscope field of view showing a metric ruler aligned across the circular viewing area, with measurement markings used to determine the diameter of the field of view.

Figure 2.3. Field of view diameter being measured in a microscope with a metric ruler. Mikael Häggström, 29 January 2020. CC0.

Arrange the slide so that the horizontal line is aligned along the midline of the field as seen above. Align one of the vertical lines so that it is at the edge of your field of view, as you can see on the right side of the picture. The space between the leading edge of the horizontal lines is 1 mm in distance, see the yellow bar.

1. Measure the distance of the field of view for the scanning lens (4X)

a. _____ mm.

b. _____ um

2. Measure the distance of the field of view for the low power lens (10x)

a. _____ mm

b. _____ um

When you use high-power and oil-immersion lenses, you should not be able to measure the field of view directly. At this point, you will have to calculate the field of view. This calculation is a simple conversion using the formula below:

Conversion Formula:

(Diameter of the field; **A**) X (Total Magnification: **A**) = (Diameter of the field; **B**) X (Total Magnification: **B**)

In this case “A” stands for the known field information of the measured field of view of either the scanning or low-power magnification and “B” stands for the information of the new or unknown lens magnification (e.g., high-power or the oil immersion lenses)

3. Figure out the distance of the field of view for the high-power objective lens (40X)

a) _____ mm.

b) _____ um

4. Determine the distance of the field of view for the oil-immersion objective lens (100X)

a) _____ mm

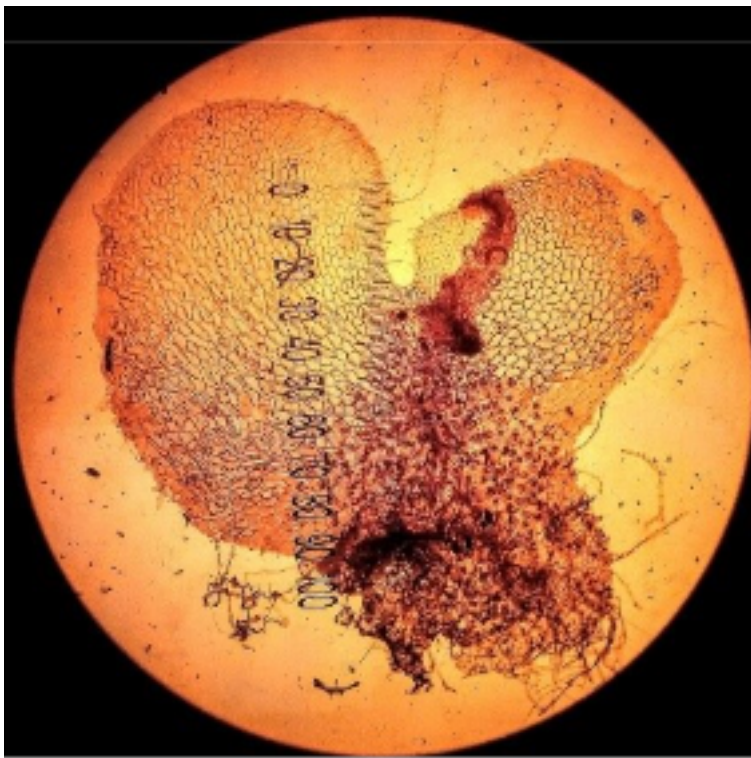
b) _____ um

Focusing Exercise(s)

- Letter “e” exercise
 - Obtain a letter “e” slide and place it on the mechanical. Note the position of the “e.” Draw and label it on the page.
 - Center the “e” and look at it through the oculars. Draw this “e” and label it.
 - Describe the differences there are between the two ‘e’s. Hint: look at and describe its position and orientation.

Field of View and Depth of Field of a Microscope

- When viewing a specimen through a microscope, the visible area appears circular and is referred to as the **field of view**. The diameter of this circle decreases as magnification increases.

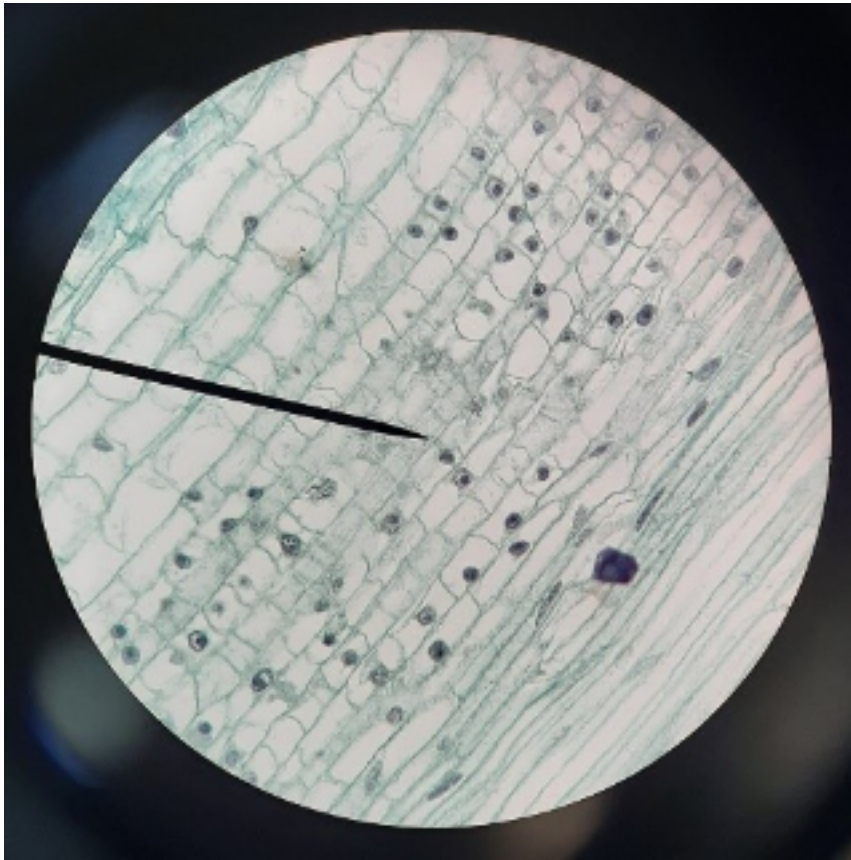


{alt="A close-up of a cell

AI-generated content may be incorrect.”}

Circular microscope field of view displaying the prothallium of *Polypodium vulgare* at 4× magnification, showing a thin, flat, green plant structure centered within the viewing area.

Figure 2.4. Circular microscope field of view showing the prothallium (prothallus) of *Polypodium vulgare*. Taken with a light microscope on x4 magnification. Image: Viséan, 2019, CC BY-SA 4.0, <https://commons.wikimedia.org>



scope view of cells under a microscope

AI-generated content may be incorrect.”}

Circular microscope field of view displaying an unknown plant specimen, showing cellular or tissue structures within a round viewing area under magnification.

Figure 2.5. A view of an unknown plant under a microscope. Palmer-Harris, L. (2021, December 28).CC BY SA 4.0.

- The **depth of field** is the vertical distance (top to bottom) within the specimen that stays in focus at one time. As magnification increases, the depth of the field decreases. Therefore, the scanning objective lens provides the greatest depth of field, while the oil immersion objective lens has the shallowest depth of field.



blue x

{alt="A close-up of a red and

AI-generated content may be incorrect.”}

Microscopic view showing three overlapping yarn fibers at 40 $\times$ magnification, with only portions of the fibers in sharp focus at a time, illustrating limited depth of field.

Figure 2.6. View showing three overlapping yarn fibers (40 $\times$). Image from Botweb, the University of Wisconsin, <https://botweb.uwsp.edu/130y.htm>.

- A prepared slide having three differently colored silk fibers or threads is commonly used to show depth of field. In this preparation, the threads are arranged to cross at a single point, with each thread positioned at a different vertical level. By focusing at 400 $\times$ and adjusting the fine focus and seeing which thread comes into focus first, second, and last, the relative positions of the threads—top, middle, and bottom—can be figured out.

2.18.1 Depth of Field Using Crossed Threads

1. Obtain a prepared slide containing three crossed threads.
2. Place the slide on the microscope stage. Using the **scanning objective lens**, bring the threads into focus with the **fine adjustment knob**. Figure out which thread is positioned on the top, in the middle, and at the bottom of the slide.
3. Switch to the **100 $\times$ (low power) objective lens**. Locate the top thread and then focus on the bottom thread. Compare the amount of fine focus adjustment needed at this magnification to that used with the scanning objective.

- Did you need to move the adjustment knob more or less? _____

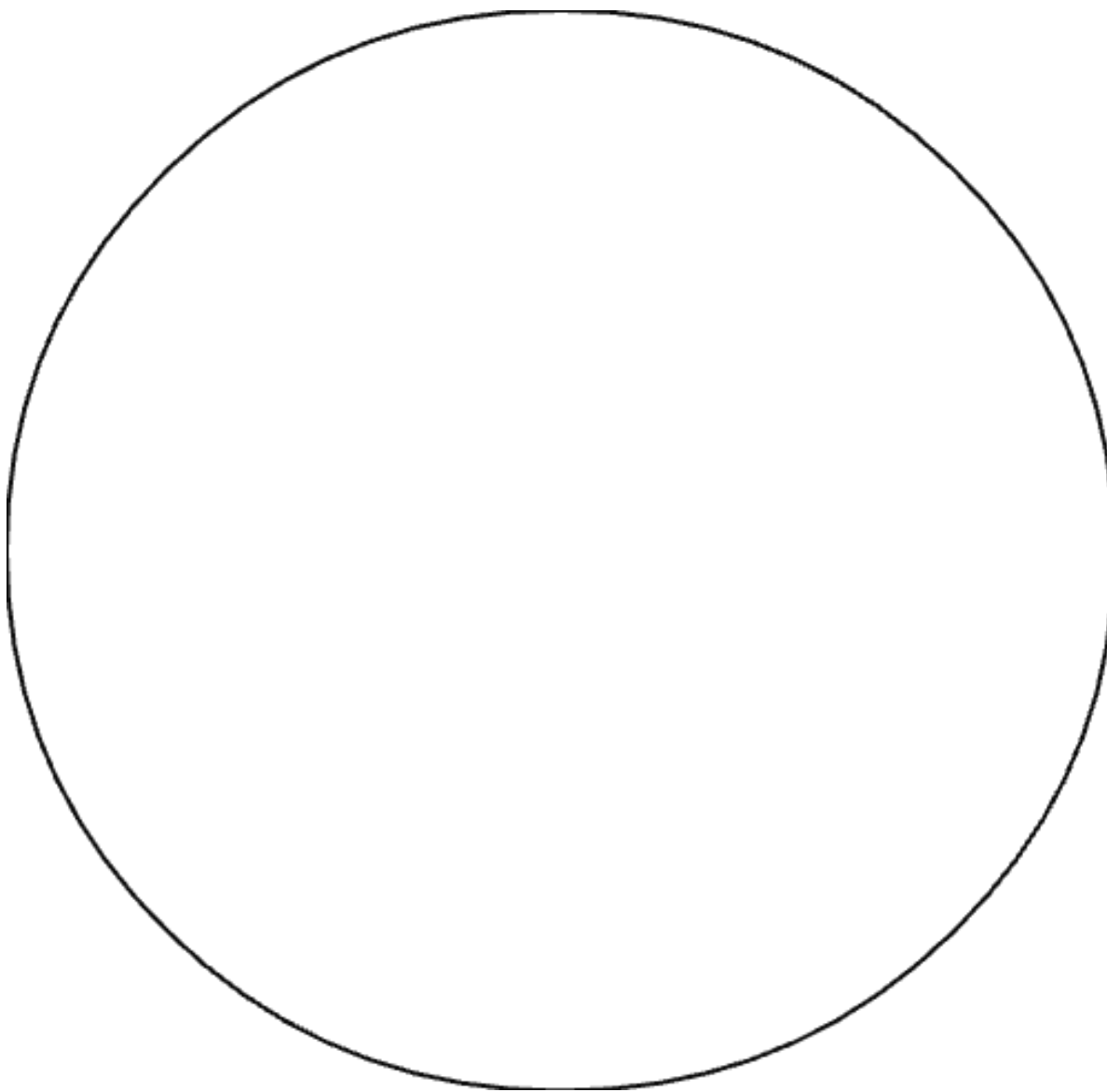
4. Change to the **400× (high power) objective lens**. At this magnification, each thread may occupy most of the field of view.
- Is the entire thickness of a thread in focus as it was under the scanning objective lens? _____
 - What does this observation say about the depth of field at high magnification? _____
5. Using only the **fine focus knob**, figure out the relative vertical position of each colored thread.
- Bottom thread: _____
 - Middle thread: _____
 - Top thread: _____

Examining Tissues

Your instructor will guide you in observing a series of prepared organism slides to practice proper microscope technique. Examine each slide using total magnifications of **40×, 100×, and 400×**.

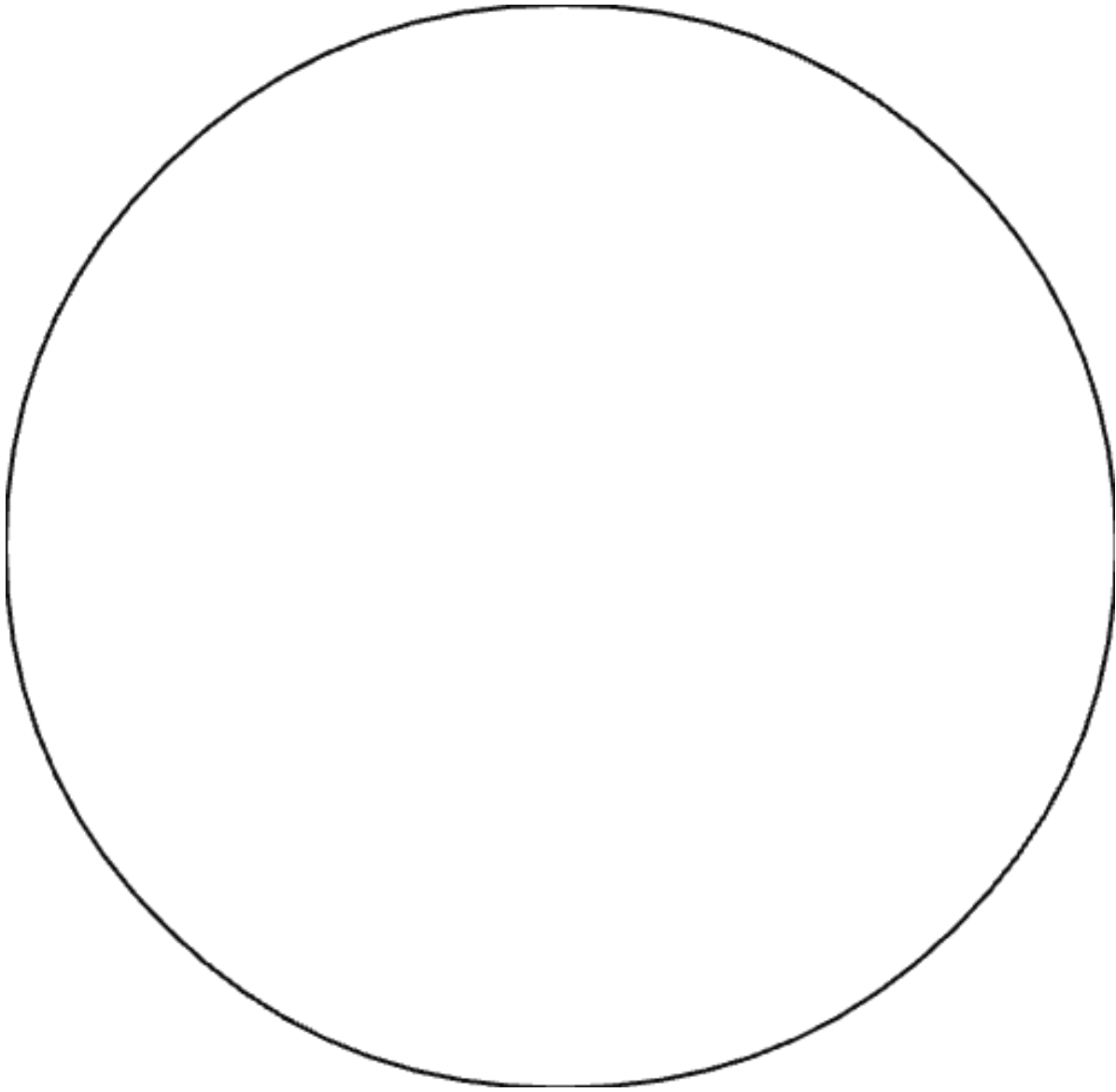
Place particular emphasis on effective focusing and observation at **400× magnification**, as this magnification will be used during the practical examination. Draw and label the organisms from the prepared slide

2.19 Prepared slide



Identification of the organism: _____

Prepared slide



Identification of the organism: _____

2.20 Preparing a Wet Mount

You will have to prepare wet mount slides of fresh specimens on many occasions during this course. You will often need to apply a biological stain to a specimen to see the structures within cells. Such stains combine with cellular components. Selective use of available stains enables the observer to see desired components. It is also important when preparing wet mount

slides to use the thinnest section of a specimen possible. You will use a scalpel or razor blade to section your specimens. A wet mount requires a layer of liquid to completely fill the space between the slide and coverslip, immersing the specimen and enabling the observer to clearly focus through the microscope.

2.21 Procedure:

I. Wet mount of an Onion Skin

Watch the video on [this](#) link to see how to prepare wet mount of an onion skin and then follow the instructions below:

1. Prepare the specimen

- Peel a small, rectangular piece of transparent epidermis (inner layer) from a sliced onion.
- Place the onion skin flat in the center of a clean glass slide.

2. Stain

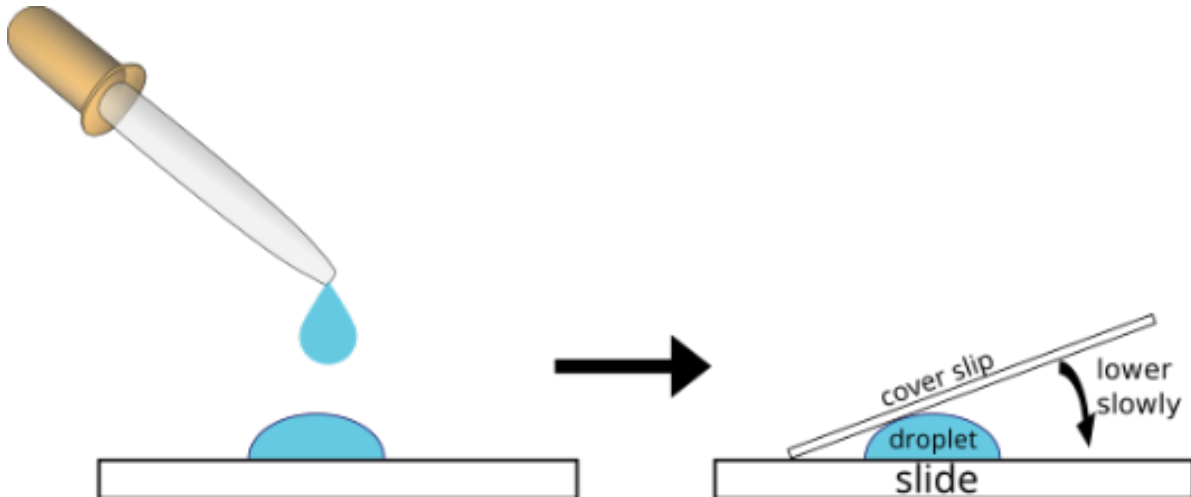
- Add one drop of methylene blue directly onto the onion skin.
- Let it sit for 1 to 2 minutes, so the stain penetrates the cells.

3. Remove excess stain and rinse

- Gently touch a corner of a paper napkin to the slide edge (not the specimen) to wick away excess stain.
- Add one small drop of water to rinse the stained area, then wick away the rinse with the napkin.
- Repeat the rinse once, if needed, remove the excess each time.

4. Mount with a coverslip

- Place one final small drop of water on the specimen.
- Hold the coverslip at a 45° angle and slowly lower it onto the drop to spread the liquid and avoid air bubbles.



A student lowers a coverslip at a 45-degree angle onto a glass slide having a drop of liquid and a specimen, allowing the liquid to spread evenly and prevent air bubbles from forming beneath the coverslip.

Figure 2.7 Image of placing a coverslip on a glass slide when making a wet mount. Image courtesy of New York City College of Technology. BY CC NC SA 3.0.

5. Clean slide edges

- Run the edge of a paper napkin around the coverslip to remove excess fluid.
- Wipe and dry the bottom of the slide so it is clean for viewing.

6. Observe under the microscope

- Secure the slide on the stage. Start with the scanning objective lens (4X) to find tissue; center a clear area in the field of view.
- Switch to low power (10X), re-center and refocus, then move to high power (40X) for details (40X objective with a 10X eyepiece = 400X).
- Always center and refocus before increasing magnification. Draw or image what you see.

7. Finish and dispose

- Rotate the scanning (4X) objective lens into place and lower the stage. Remove the slide and place it in the “Used Slides” container.
- Clean and store equipment per lab protocol and wash your hands.

Safety notes: avoid pressing the coverslip, handle methylene blue carefully, and follow your lab’s disposal procedures.

II. Wet mount of human cheek cells

Watch the video on [this link](#) to see how to prepare wet mount of a cheek cell and then follow the instructions below:

1. Wash your hands with soap and water.
2. Watch the onionskin wet mount video first.
3. Use a sterile cotton swab or the flat end of a clean toothpick to gently scrape the inside of your cheek (do not poke or scrape hard).
4. Place a clean microscope slide on a flat surface and add one drop of water to the center.
5. Rub the swab or toothpick tip gently through the drop for ~4 seconds to transfer cells. Discard the used swab/toothpick safely.
6. Add one drop of methylene blue to the smear (at the edge of the drop to avoid bubbles).
7. Carefully lower a cover slip at an angle over the drop to avoid trapping air bubbles.
8. Blot any excess stain from the slide edge with absorbent paper.
9. Place the slide on the microscope stage. Start with the scanning (4X) lens to locate the sample, then switch to a 40X objective (400X total with a 10X eyepiece) to observe stained cheek cells.
10. When finished, dispose of the cover slip and slide per your lab's biohazard or sharps protocol, and wash your hands.

Safety notes: wear gloves and eye protection if available; avoid swallowing stains; do not reuse contaminated toothpicks; follow institutional disposal rules.

2.22 II. Wet Mount Preparation of Cultured Organisms and Pond Water

View the [instructional video](#) in this link to review wet mount preparation techniques using cultured organisms. Then complete the following procedure.

1. Gather materials: plastic pipette, clean slide, coverslip, and your sample (cultured organisms or pond water).
2. Use the pipette to draw liquid from near the bottom of the container (where organisms collect).
3. Put 1 or 2 drops of the sample in the center of the slide.
4. Hold the coverslip at a 45° angle and slowly lower it onto the drop to avoid air bubbles.

5. Blot excess liquid by touching a paper napkin to the coverslip edge; dry the underside of the slide.
6. Place the slide on the stage. Start with the 4× (scanning) objective lens to find the specimen, center it, then move to 10× and 40×—recenter and refocus at each step.
7. Record observations with a labeled drawing.
8. Finish: return the 4× objective lens, lower the stage, remove the slide, and put it in the “Used Slides” container.

Safety: avoid contaminating the sample, do not a pipette by mouth, and follow lab disposal rules.

2.23 End-of-Lab Procedures

- Clean and return all prepared slides to their designated storage trays.
- Turn off and unplug the microscope; neatly wrap the power cord around the cord holder or base.
- Rotate the head to face inward toward the microscope body.
- Set the objective turret to the scanning (4×) objective lens.
- Lower the stage fully.
- Place the dust cover over the microscope (if available).
- Return the microscope to its assigned storage cabinet.

Troubleshooting Guide

1. Image too dark
 - Check power: ensure the microscope is plugged in and switched on; confirm the light is lit.
 - Increase light: raise the rheostat (brightness control).
 - Open the iris diaphragm and/or raise the condenser to let more light through.
 - Fine-tune the rheostat until the image is bright and white.
2. Unable to see a single image with both eyes
 - Move your head slightly away from the eyepieces.

- Adjust interpupillary distance by sliding the oculars together or apart until one image appears.
3. Stationary spot in the field of view
 - The lens is likely to be dirty. Use only lens paper to gently clean the oculars and objective lenses.
 - If needed, carefully remove the ocular to clean the inner surface.
 - Do not use tissues, paper towels, or clothing (they can scratch lenses).
 4. Eyelashes or other near eye artifacts visible
 - e too close to the eyepieces—move your head back a little.
 5. Nothing visible at high power
 - Always focus on scanning ($4\times$) and low ($10\times$) first, center the specimen, then switch to high ($40\times$).
 - Make sure the specimen is centered before increasing magnification.
 6. Only part of the field is illuminated (half-moon)
 - Ensure the objective clicks fully into place on the nosepiece.
 - Adjust the condenser and its centering to align the light.
 7. Image stays blurry and will not focus
 - Check the slide orientation (not upside down) and that it is secured in the stage clips.
 - Recenter the specimen and refocus at lower magnification before switching up.
 8. Eye strain or headache from using the microscope
 - Focus first with your dominant eye (e.g., right eye), then adjust the other ocular diopter.
 - Set interpupillary distance correctly and reduce light intensity if it is too bright.
 - Take short breaks to rest your eyes.

Quick maintenance/reminders

- Always start with the lowest magnification, center the specimen, then increase power.
- Use only lens paper for optics.
- If problems persist, notify the instructor or lab technician.

2.24 Student Post-Laboratory Policy

Students must complete all assigned post-laboratory questions, including both short-answer and multiple-choice items, following the completion of the laboratory exercise. These questions are designed to assess understanding of key concepts and proper application of laboratory techniques.

All responses must be completed prior to leaving the laboratory unless otherwise directed by the instructor. Students must present their completed work to the instructor for review and verification before departure.

Be advised that completion of the post-laboratory questions may be required for a grade at the discretion of the instructor. Failure to complete, submit, or obtain instructor for review of the assigned questions may result in a loss of credit or the assignment being considered incomplete.

It is the responsibility of each student to ensure compliance with all post-laboratory requirements and grading expectations as outlined by their instructor.

2.25 Student Post-Laboratory Questions

(To be completed after the laboratory to assess comprehension)

1. How did the depth of field change as you increased magnification from 4× to 40×?
2. In the crossed threads slide, how were you able to determine which thread was on top, middle, and bottom?
3. Why was it more difficult to keep the entire specimen in focus at higher magnifications?
4. Describe one structure you observed in the wet mount that was not visible at lower magnification.
5. What mistakes could lead to a broken slide or damaged objective lens, and how can they be avoided?
6. How does proper microscope care extend the life of the instrument and improve data quality?

Multiple choice questions

Directions: Choose the best answer for each question.

Basic Knowledge and Application

1. Which part of the microscope is closest to the eye?
 - A. Objective lens
 - B. Condenser
 - C. Ocular lens
 - D. Stage
2. What is the total magnification if the ocular lens is $10\times$ and the objective lens is $40\times$?
 - A. $4\times$
 - B. $50\times$
 - C. $400\times$
 - D. $100\times$
3. Which objective lens should always be used first when focusing on a specimen?
 - A. Oil immersion
 - B. High power
 - C. Low power
 - D. Scanning
4. What is the function of the coarse focus knob?
 - A. Adjust light intensity
 - B. Make small adjustments to focus
 - C. Move the stage for large focusing adjustments
 - D. Change magnification
5. Which part controls the amount of light reaching the specimen?
 - A. Condenser
 - B. Iris diaphragm
 - C. Objective lens
 - D. Mechanical stage
6. What happens to the field of view as magnification increases?
 - A. It increases
 - B. It stays the same
 - C. It decreases
 - D. It becomes brighter
7. Which structure holds the objective lenses and rotates to change magnification?
 - A. Arm
 - B. Base
 - C. Revolving nosepiece
 - D. Stage clips
8. What is the correct way to carry a microscope?
 - A. With one hand on the base
 - B. With one hand on the arm

- C. With two hands (arm and base)
 - D. By holding the eyepiece
9. Which lens requires the use of immersion oil?
- A. Scanning lens
 - B. Low power lens
 - C. High power lens
 - D. Oil immersion lens
10. What is resolution?
- A. The brightness of an image
 - B. The amount of light entering the microscope
 - C. The ability to distinguish two close points as separate
 - D. The total magnification

2.26 Higher-Order Thinking (Analysis & Application)

1. A student switches directly from scanning ($4\times$) to high power ($40\times$) and cannot find the specimen. What is the most likely reason?
- A. The light intensity is too high
 - B. The specimen was not centered before increasing magnification
 - C. The objective lens is broken
 - D. The eyepiece needs cleaning
2. If an image is too dark, which combination of actions would best improve visibility?
- A. Lower the stage and use coarse focus
 - B. Increase illumination and open the diaphragm
 - C. Switch to higher magnification
 - D. Close the diaphragm and turn off the light
3. Why does depth of field decrease as magnification increases?
- A. The lens becomes larger
 - B. More light passes through the specimen
 - C. A thinner section of the specimen can be kept in focus
 - D. The field of view increases
4. Two students measure the same specimen. One gets consistent results that are slightly off from the true value, while the other gets varied results around the correct value. Which is true?
- A. First student is accurate, second is precise
 - B. First student is precise, second is accurate
 - C. Both are equally accurate
 - D. Neither is precise

5. If a specimen moves to the left when viewed under the microscope, what direction was the slide actually moved?
 - A. Left
 - B. Right
 - C. Up
 - D. Down
6. A student uses the coarse focus knob while viewing a slide under high power and breaks the slide. What mistake did the student make?
 - A. Used the wrong lens
 - B. Turned the light off
 - C. Used coarse focus at high magnification
 - D. Did not center the specimen
7. A biologist needs to observe small internal cell structures. Which combination of features is most important?
 - A. Low magnification and large field of view
 - B. High magnification and high resolution
 - C. Low resolution and high brightness
 - D. Large stage and low light
8. Why is it important to center the specimen before switching to a higher magnification?
 - A. It improves light intensity
 - B. It prevents lens damage
 - C. The field of view becomes smaller at higher magnification
 - D. It increases resolution
9. A student notices that only half of the field of view is illuminated. What is the most probable cause?
 - A. Dirty slide
 - B. Objective lens not fully clicked into place
 - C. Stage is too low
 - D. Specimen is too thick
10. Why is starting with the scanning objective considered best practice?
 - A. It gives the highest resolution
 - B. It allows faster focusing with fine adjustment
 - C. It provides the widest field of view and reduces the risk of damage
 - D. It requires less light

2.27 Student resources:

2.28 Microscopy

2.29 The Light Microscope. A review of lab materials.

<https://openlab.citytech.cuny.edu/bio-oer/biology-basics/microscopy/>.

2.29.1 Learn the parts of the microscope (activity)

- a. Visit <https://www.ncbionetwork.org/iet/microscope/>
 - i. Click on “Explore”
 - ii. Click on parts “?” to learn their name and function

Table 2.2. Chapter 2: Microscopy Glossary of Terms and Definitions

This table provides key terminology related to microscopy, along with concise definitions of each term. It serves as a reference for understanding the structure, function, and proper use of the microscope, as well as concepts such as magnification, and resolution.

Chapter 2: Microscopy Glossary	
Terms	Definitions
binocular	a microscope having two eyepieces
brightfield microscope,	a compound microscope with two or more lenses that produce a dark image on a bright background.
coarse focusing knob	used for large-scale movements with 4 and 10 objective lenses
condenser lens	(located below the stage), which focuses all the light rays on the specimen to maximize illumination.
fine focusing knob	used for small-scale movements, especially with 40 or 100 objective lenses.
fixation	The process of attaching cells to a slide.
illuminator	provides light for the brightfield microscope
lens	a medium with a curved surface that refracts and focuses on light to produce an image.
magnification	the ability of a lens to enlarge the image of an object when compared to the real object
monocular microscope	a microscope having a single ocular lens

Chapter 2: Microscopy Glossary

objective lenses	the lenses far end of the body tube
ocular lens	the eyepiece of a microscope
oil immersion lens,	a special lens designed to be used with immersion oils.
refraction	occurs when light waves change speed and direction as they pass from one medium to another.
resolution	The ability to tell that two separate points or objects are separate.
rotating nose-piece, also known as the revolving turret, stage	sits below the head of the microscope and locks the objective lens into position over the stage aperture by rotating in either direction.
Total magnification	(a platform) of the microscope total magnification is the product of the ocular magnification times the objective (ocular magnification $\times$ objective magnification)
wet mount	slide preparation in which the specimen is placed on the slide in a drop of liquid.
x-y mechanical stage knobs.	These knobs move the slide on the surface of the stage, but do not raise or lower the stage.

Citations

1. Figure 2.1. Components of a typical bright field microscope. GCG(jawp). 2008. https://commons.wikimedia.org/wiki/File:Optical_microscope_nikon_alphaphot.jpg#/media/File:Optical_microscope_nikon_alphaphot.jpg.
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3. Figure 2.3: Field of view diameter measured in a microscope with a metric ruler. Hägström, M. (2020, January 29). Field of view diameter being measured in a microscope with a metric ruler [Photograph]. Wikimedia Commons. https://commons.wikimedia.org/wiki/File:Field_of_view_diameter_in_microscopy.jpg (CC0)
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3 Molecules, Water, and pH

3.1 Pre-Lab Activity 3.1: Questions

Objectives

1. Represent molecules using molecular models
2. Write a molecular formula based on a structural formula
3. Identify and demonstrate various properties of water that make it essential for life.
4. Identify acids and bases and learn how to measure their pH
5. Identify the role of buffers in living organisms

Multiple Choice (10)

1. Which of the following is an inorganic compound?
 - A. Glucose
 - B. Protein
 - C. Water
 - D. Lipid
2. Organic molecules always contain:
 - A. Nitrogen
 - B. Carbon and hydrogen
 - C. Oxygen only
 - D. Sulfur
3. Carbon can form how many covalent bonds?
 - A. 2
 - B. 3
 - C. 4
 - D. 6
4. A molecular formula shows:
 - A. Exact bonding structure
 - B. Atom arrangement
 - C. Types and numbers of atoms
 - D. Bond angles

5. What type of bond involves shared electrons?
 - A. Ionic
 - B. Hydrogen
 - C. Covalent
 - D. Metallic
6. Isomers are molecules that:
 - A. Have different formulas
 - B. Have same atoms but different arrangements
 - C. Are always inorganic
 - D. Are always identical
7. Water is considered polar because:
 - A. It has equal charge distribution
 - B. It contains hydrogen bonds only
 - C. Electrons are shared evenly
 - D. Electrons are shared unevenly
8. pH measures:
 - A. Oxygen levels
 - B. Hydrogen ion concentration
 - C. Temperature
 - D. Salt concentration
9. A buffer functions to:
 - A. Increase pH rapidly
 - B. Neutralize all acids
 - C. Resist changes in pH
 - D. Eliminate hydrogen ions

Short Answer (10)

11. Define an organic molecule.
12. What is a covalent bond?
13. What is the difference between a molecular and structural formula?
14. What does the formula $\text{C}_2\text{H}_6\text{O}$ tell you about a molecule?
15. What is a functional group?
16. Explain what an isomer is.
17. Why is water considered an excellent solvent?
18. Define cohesion and adhesion.
19. What is surface tension?

20. Why is pH important in living organisms?

Key Terms to Define

- Organic compound
- Inorganic compound
- Covalent bond
- Molecular formula
- Structural formula
- Abbreviated structural formula
- Isomer
- Polarity
- pH
- Acid
- Base
- Buffer

3.2 Introduction: Molecules, Water, and pH

Biology is the study of life, and understanding life begins with understanding the chemistry that makes it possible. In this first-semester biology lab, you will explore the fundamental chemical principles that underlie biological systems, preparing you for advanced work as a biology major or pre-med student. Through hands-on experiments and model-based activities, you will connect basic chemical concepts to real biological function.

Chemical compounds can be divided into two broad categories: **inorganic** and **organic**. Inorganic compounds, such as water, salts, acids, and bases, generally do not contain carbon–hydrogen (C–H) bonds and play essential roles in maintaining life processes. Organic molecules, on the other hand, are defined by the presence of carbon and C–H bonds. Carbon’s unique ability to form four covalent bonds allows it to create a wide variety of complex molecules, especially when combined with other key elements such as oxygen, nitrogen, phosphorus, and sulfur—elements that make up the vast majority of living matter.

In this lab, you will investigate how atoms form **molecules** through **covalent bonding**, where electrons are shared between atoms. You will learn how to represent

molecules using **molecular formulas**, **structural formulas**, and **abbreviated structural formulas**, each providing different levels of detail. Using molecular model kits, you will build three-dimensional representations of molecules to better understand how atomic arrangement influences molecular shape and function. You will also explore **isomers**, which are molecules with the same molecular formula but different structures, demonstrating how small differences in arrangement can lead to significant biological differences.

A major focus of this lab is **water**, one of the most important inorganic molecules for life. Because of its polar structure, water exhibits unique properties such as cohesion, adhesion, high surface tension, and high specific heat. These properties allow water to act as a solvent, regulate temperature, and move through narrow spaces via capillary action—all essential for sustaining life. You will perform experiments to observe and measure these properties directly.

You will also examine **acids, bases, and pH**, which are critical for maintaining proper conditions in biological systems. The pH scale measures the concentration of hydrogen ions in a solution and determines whether a solution is acidic, neutral, or basic. Because the pH scale is logarithmic, even small changes represent significant shifts in chemical conditions. Living systems depend on tightly regulated pH levels—for example, human blood must remain within a narrow range to support proper enzyme and cellular function. To maintain this balance, organisms rely on **buffers**, which resist changes in pH when acids or bases are introduced.

By combining theory with experimentation, this lab will help you develop essential scientific skills, including observation, data collection, analysis, and critical thinking. These foundational concepts will support your future studies in biology, chemistry, and medicine, and provide insight into how chemical processes sustain life at every level.

3.3 Activity 1: Lab Experiment

Watch the video on the link [here](#) to understand the activity we will be performing.

Objectives:

By the end of this laboratory exercise, students should be able to:

1. Describe how the addition of acids and bases affects the pH of a solution.
2. Demonstrate proper use of a pH probe and computer interface to measure pH.
3. Accurately collect, record, and organize experimental data.

4. Construct and interpret graphs showing pH changes in response to added substances.
5. Compare pH changes in plain water versus buffered solutions.
6. Explain how buffers resist changes in pH when acids or bases are added.
7. Analyze experimental results to identify patterns and draw conclusions.
8. Relate the concept of buffering to biological systems and homeostasis.

Materials

- 1 laptop computer
- 1 computer interface
- 2 pH probes
- 2 microstirrers (*rinse with deionized water before use*)
- Stir plate setup (stir plate, rod, and clamp)
- 1 – 250 mL beaker (for rinsing probes)
- 2 – 50 mL beakers (*rinse before each use*)
- 1 – 100 mL graduated cylinder (*rinse before use*)
- ~125 mL tap water (for probe storage)
- 1 rinse bottle with distilled/deionized water
- 0.10 M HCl (acid) with dropper
- 0.10 M NaOH (base) with dropper
- 2 Bufferin tablets

General Setup

1. Put on your **safety goggles**.
2. Add **40 mL of distilled water** to each of two 50 mL beakers.
3. Label the beakers:
 - **A (Acid)**
 - **B (Base)**
4. Place both beakers on stir plates.

Preparing the pH Probes

5. Remove each pH probe from its storage container.
6. Rinse the probe tip thoroughly:
 - Hold it over a 250 mL beaker
 - Use distilled water to rinse

Important:

- Do **not** let the probe dry out
- Keep it in tap water when not in use
- The tip is **fragile glass—handle carefully**

Computer Setup

7. Set up the computer and interface (follow your instructor's guide).
8. Plug the probes into:
 - **CH1**
 - **CH2**
9. Open the “**Bio 110**” folder on the desktop.
10. Open “**Acids Bases Template.**”

You should see both probes listed on the screen. If not, check connections or try a different USB port.

Part 1: Testing Acids and Bases in Water

1. Place the probes into the beakers:
 - **CH1 probe → Beaker A (acid)**
 - **CH2 probe → Beaker B (base)**
2. Turn on the stir plates.
3. Click “**COLLECT**” to begin recording pH.
4. Wait until the pH readings stabilize.
5. Click “**KEEP,**” enter **0** (no drops yet), and click OK.

Adding Acid and Base

6. Add:

- 5 drops of acid to Beaker A
 - 5 drops of base to Beaker B
7. Wait for the readings to stabilize.
 8. Click “**KEEP**” and enter the **total number of drops added**.
 9. Repeat steps 6–8 until you reach **30 total drops**.
 10. Click “**STOP**.”
 11. Save your data (**Experiment** → **Store Latest Run**).
 12. Record your results in **Table 1**.

Table 1: Water (No Buffer)

Total Drops Added	pH (Acid - Beaker A)	pH (Base - Beaker B)
0		
5		
10		
15		
20		
25		
30		

Graph:

Graph the data you collected.

Title:

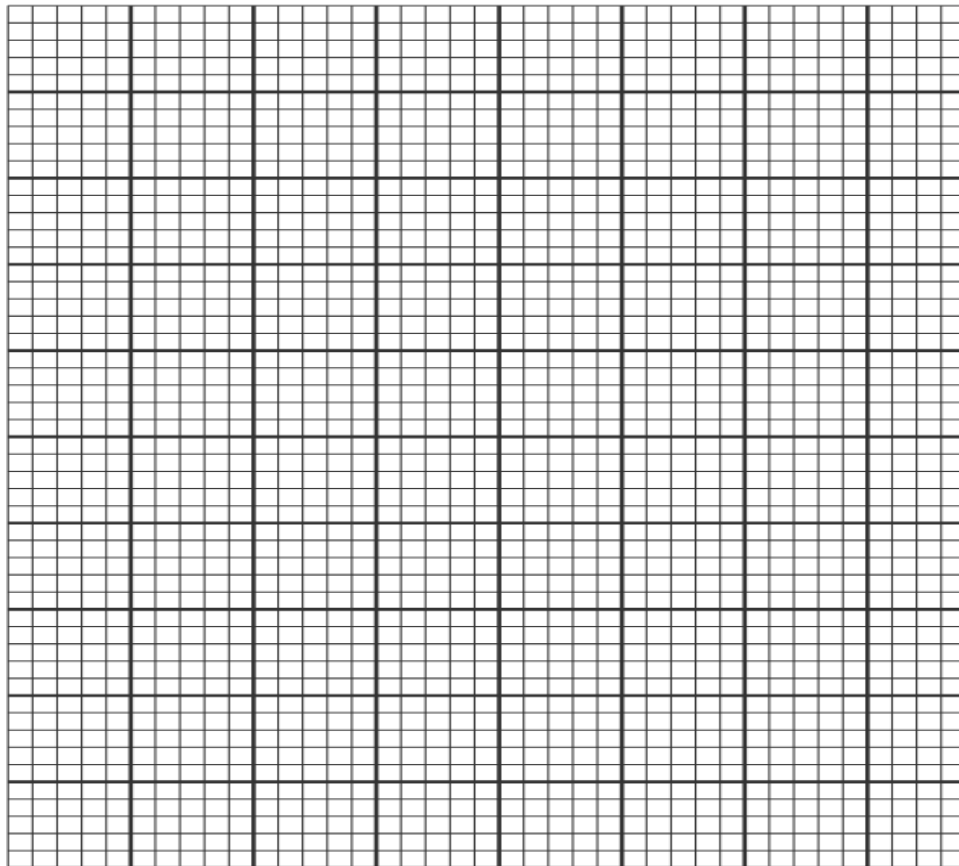


Figure 3.1: A graph paper with a grid AI-generated content may be incorrect.

After Part 1

13. Rinse probes and place them in tap water.
14. Rinse both beakers thoroughly with deionized water.

Part 2: Testing Buffers (Bufferin)

1. Add **40 mL of distilled water** to two clean 50 mL beakers.
2. Place **one Bufferin tablet** in each beaker and let it dissolve.
3. Repeat the same procedure as in Part 1 (steps 1–13).

Table 2: Buffered Solution (Bufferin)

Total Drops Added	pH (Acid - Beaker A)	pH (Base - Beaker B)
0		
5		
10		
15		
20		
25		
30		

Graph:

Graph the data you collected.

Title:

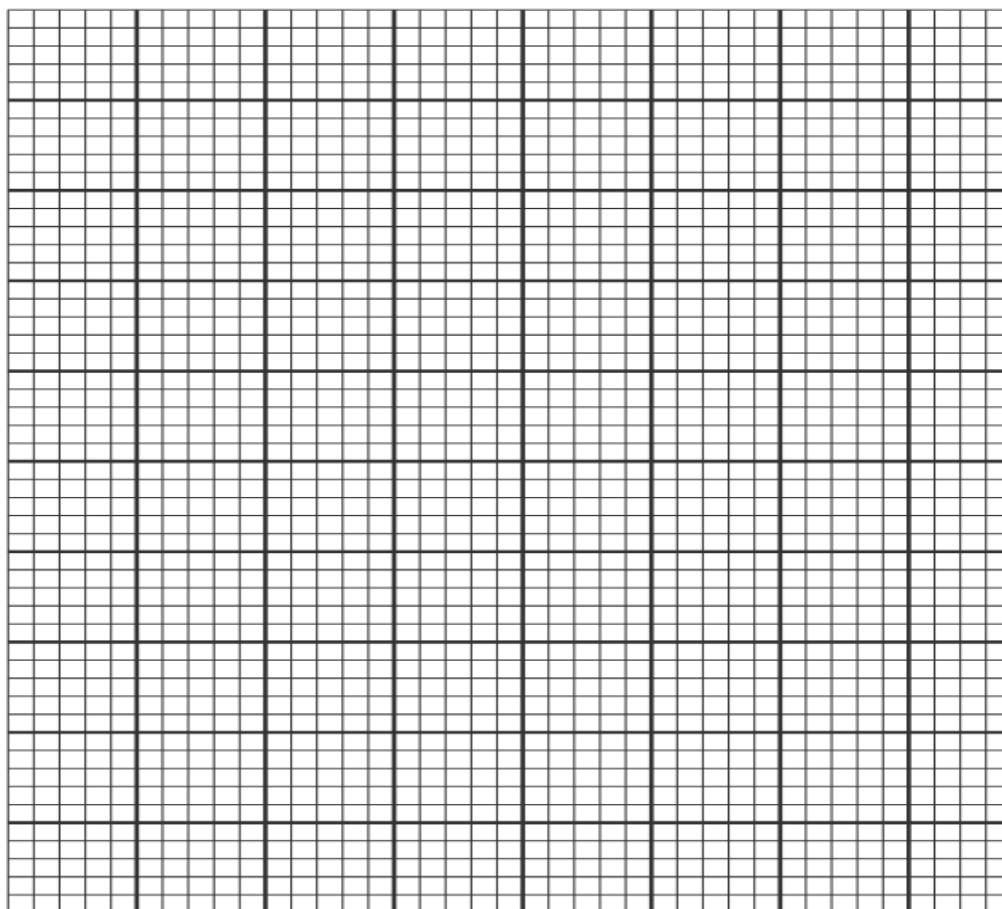


Figure 3.2: A graph paper with a grid AI-generated content may be incorrect.

Clean-Up

1. Disconnect the pH probes.
2. Rinse them and return them to storage solution.
3. Clean all glassware (especially the 50 mL beakers).

3.4 Activity 2: Demonstration of Capillary Action (Adhesion and Cohesion)

In this exercise, you will observe how water can move from one container to another without direct pouring. This activity demonstrates the properties of water such as cohesion and adhesion, which allow water to travel through materials like paper towels. Over time, you will record your observations to better understand how water moves through capillary action.

1. Take two 250 ml beakers and put them side by side.
2. Fill one with water. Keep the second empty.
3. Take a piece of paper towel and fold it lengthwise in the middle a few times to get a thin strip. Fold it in the middle to make a V shape.
4. Dip one end of the paper towel strip into the empty beaker and the other end into the beaker full of water. Allow this to sit until the end of the lab period. Check the beakers every 60 mins and make observations. Record your observations below.

Table 3 (Paper Towel Water Movement Observations):

A table for recording observations during an experiment where a paper towel connects two beakers, one with water and one empty. Students note changes over time, such as water movement and levels in each beaker, at regular intervals.

Elapsed Time (in minutes)	Observations
0	
60	
120	

3.5 Activity 3: Demonstration of Surface Tension (Adhesion and Cohesion)

In this exercise, you will investigate the property of surface tension in water. Surface tension is caused by hydrogen bonding between water molecules, creating a thin “skin” at the surface. By observing how a paperclip floats and how detergent affects this behavior, you will better understand how intermolecular forces influence the properties of water.

1. Fill a culture dish halfway with water. Float a small paperclip on the surface of the water. This requires careful placement of the paperclips to avoid breaking the surface tension of the water.
2. Use a dropper to carefully deposit a drop of water near the floating paperclips. Record your observations.
3. Next, deposit several drops of liquid detergent near the floating paperclips. After applying the detergent, try to float another paperclip. The detergent disrupts the hydrogen bonds in water. Record your observations below.

Table 4 (Detergent and Surface Tension Observations):

A table for recording observations during an experiment where detergent is added to water with floating paperclips. Students note changes after the detergent disrupts hydrogen bonding, such as paperclips sinking and differences in surface tension.

Drop- per con- tents	Did it cause the paperclip to sink?	Did this substance break the surface tension?
Water		
Deter- gent		

3.6 Activity 4: Building Models

In this activity, you will explore how molecules are represented using molecular, structural, and abbreviated structural formulas. A molecular formula, such as glycine (C H N O), shows the number and types of atoms present, while structural formulas display how those atoms are bonded. Abbreviated structural formulas simplify these representations by grouping atoms, making complex molecules easier to visualize.

You will use a molecular model kit to build three-dimensional representations of molecules, helping you understand how atoms connect through covalent bonds. As you construct models, you will also observe how the same molecular formula can form different structures, known as isomers. Additionally, you will examine how chemical reactions involve breaking and forming covalent bonds to create new products.

1. Look at the four most common spheres and fill out which element they represent and how many bonds they can make in the table below.

Table 5 (Element Spheres and Bonding):

A table for identifying common elements shown as spheres in molecular models, with columns to list each element and the number of bonds it typically forms.

Sphere color	Element	# of bonds it forms
black		
white		
red		
blue		

2. Build a molecular model for each molecule listed in the table below. Use the molecular formula to collect the correct spheres first, then put them together so that every hole is filled with a bond. Have your instructor verify your models. Write the structural formula from the model you built in the table below. *Hint: some of these molecules may include double or triple bonds*

Table 6 (Molecular Model Building Activity):

A table for recording molecular model results, where students build each molecule using spheres and bonds, then write the corresponding structural formula based on the completed model.

Molecule	Molec- u- lar For- mula	Struc- tural For- mula
Methane		
Ethane	CH	
Ethene	C H	
Methanol	C H	
	CH O	

Molecule	Molecular Formula	Structural Formula
Formaldehyde		
Acetone (<i>contains a carbonyl group at a ketone position between two methyl groups</i>)	CH ₃ COCH ₃	
Glycine (a 2-carbon compound; contains an amino and a carboxyl group on each end)	C ₂ H ₅ NO ₂	

Several functional groups are commonly found in organic molecules. Using the table below, build a molecular model for each listed functional group. Then, record the corresponding structural formula for each model you construct. Refer to pages 47–49 of your textbook for guidance on the shapes and characteristic properties of these functional groups.

Table 7 (Functional Groups Molecular Models):

A table for recording common functional groups in organic molecules, where students build each group using molecular models and then write the corresponding structural formula, with dashes indicating bonds attached to a carbon atom.

Functional Group	Structural Formula	Characteristics
Amino (-NH ₂)		
Carbonyl (-CO-)		
Carboxyl (-COOH)		
Hydroxyl (-OH)		
Methyl (-CH ₃)		

- Determine the molecular formula of each molecule using the abbreviated structural formula below. Ethanol is given as an example. You do not need to build these molecules. (*remember to follow the rules of writing molecular formulas from the introduction*)

Table 8 (Determining Molecular Formulas):

A table where students determine and record the molecular formulas of molecules based on given abbreviated structural formulas, with ethanol provided as an example.

Abbreviated structural formula	Molecular formula	Abbreviated structural formula	Molecular formula
CH ₃ -CH ₂ -OH	C ₂ H ₆ O		

4. Create two different models which both have the molecular formula C₂H₆O. Draw the structural formula of each in the table below.

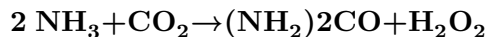
Table 9 (Isomers of C₂H₆O):

A table where students draw and compare two different structural formulas for molecules with the same molecular formula (C₂H₆O), illustrating that the same formula can produce different structures (isomers).

Molecule	Molecular Formula	Structural formula
Ethanol (<i>Start by making a hydroxyl group.</i>)		
Dimethyl ether (<i>Start by making two methyl groups</i>)		
<i>Hint: from the name of this molecule, which functional group must this molecule have?</i>		

Chemical reactions involve the breaking and forming of bonds. In this exercise, you will use molecular models to simulate these processes by modeling the formation of urea from ammonia

The overall **balanced equation** for the formation of urea from ammonia is:



Follow the steps below to construct each molecule, determine its structural formula, and identify whether it is a reactant or a product in the reaction.

- Build two ammonia (NH₃) molecules and one carbon dioxide (CO₂) molecule.
- Take one hydrogen (H) from each NH₃ and set them aside, these will be part of water.
- Take one oxygen (O), and two rods from the CO₂ model, combine these with the H from step b to make water.
- Combine the remaining molecules to construct Urea.

Table 10 (Reactants and Products—Modeling Reaction):

A table for recording molecular structures and identifying whether each is a reactant or product in a model activity. Students build ammonia and carbon dioxide molecules, rearrange atoms to form water, and combine remaining atoms to form urea, then record the structural formulas and classify each substance accordingly.

Molecule	Product or Reactant?	Structural formula
Ammonia		
Carbon dioxide		
Urea		
Water		

3.7 Post-Lab Activities

Post Lab Questions (20 Total)

1. What evidence did you observe for capillary action?
2. How did water move between the beakers and why?
3. What property of water allows paperclips to float?
4. What happened when detergent was added? Why?
5. Describe how hydrogen bonding affects water properties.

6. Compare cohesion and adhesion using your observations.
7. Explain why surface tension decreases with detergent.
8. How did adding water affect pH values?
9. Why does dilution change pH?
10. What role do buffers play in pH stability?
11. Compare experimental vs simulated pH observations.
12. What challenges did you encounter using molecular models?
13. How did model building help your understanding?
14. What differences did you observe between isomers?
15. How do structural formulas improve understanding over molecular formulas?
16. Describe one real-life application of surface tension.
17. Why is water essential for life based on lab observations?
18. How do chemical properties affect biological systems?

Check Your Understanding (Concept Questions)

1. Why are carbon-based molecules so diverse?
2. How does molecular shape impact function?
3. Why is polarity critical for biological systems?
4. What would happen if water had low surface tension?
5. Why is hydrogen bonding essential in biology?
6. Why must blood pH stay within a narrow range?
7. How do buffers prevent drastic pH changes?
8. What is the significance of isomers in biology?
9. How do structure and function relate in molecules?
10. Why are covalent bonds important in organic chemistry?

3.8 Instructor Ancillaries

Pre-Lab Support

- Provide a short mini-lecture (~10–15 min) on:
 - Bonding and molecular representation
 - Water properties
 - pH basics
- Use visual models or slides for:
 - Molecular vs structural formulas
 - Polarity diagrams
- Assign pre-lab quiz for completion grade
- Encourage students to review key terms beforehand

During Lab

- Demonstrate:
 - Floating paperclip technique
 - Proper model kit use
 - pH probe/calibration (if applicable)
- Circulate and check:
 - Model accuracy
 - Data recording
- Prompt critical thinking with questions like:
 - “What do you think will happen next?”
 - “Why did that change occur?”

Post-Lab Support

- Facilitate discussion:
 - Compare group results
 - Address misconceptions

- Provide answer key or rubric
- Connect to real-world applications:
 - Medicine (blood pH)
 - Environmental science (acid rain)

Assessment Suggestions

- Pre-lab quiz (graded)
- Lab worksheet completion
- Post-lab reflection or report
- Concept quiz (short answer emphasis)

Common Student Challenges

- Confusing molecular vs structural formulas
- Difficulty visualizing 3D structures
- Misunderstanding pH scale (logarithmic nature)
- Misinterpreting cause of surface tension changes

Tips for Success

- Use **hands-on demonstrations**
- Reinforce **connections to human physiology** (for pre-med students)
- Encourage **group collaboration**
- Provide **guided questions rather than answers**